



Reviews of Geophysics

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Key Points:

- Volcanic Ballistic Projectiles (VBPs) in volcanic deposits, theory, and direct observations are reviewed
- High-speed imaging and measurements of VBPs spinning, deforming, fragmenting, colliding, and impacting with the ground are provided
- In-flight fragmentation, collisions, and spinning are important for VBPs dynamics, and apparent drag coefficient can be higher than expected

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In-flight dynamics of volcanic ballistic projectiles

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Abstract Centimeter to meter-sized volcanic ballistic projectiles from explosive eruptions jeopardize people and properties kilometers from the volcano, but they also provide information about the past eruptions. Traditionally, projectile trajectory is modeled using simplified ballistic theory, accounting for gravity and drag forces only and assuming simply shaped projectiles free moving through air. Recently, collisions between projectiles and interactions with plumes are starting to be considered. Besides theory, experimental studies and field mapping have so far dominated volcanic projectile research, with only limited observations. High-speed, high-definition imaging now offers a new spatial and temporal scale of observation that we use to illuminate projectile dynamics. In-flight collisions commonly affect the size, shape, trajectory, and rotation of projectiles according to both projectile nature (ductile bomb versus brittle block) and the location and timing of collisions. These, in turn, are controlled by ejection pulses occurring at the vent. In-flight tearing and fragmentation characterize large bombs, which often break on landing, both factors concurring to decrease the average grain size of the resulting deposits. Complex rotation and spinning are ubiquitous features of projectiles, and the related Magnus effect may deviate projectile trajectory by tens of degrees. A new relationship is derived, linking projectile velocity and size with the size of the resulting impact crater. Finally, apparent drag coefficient values, obtained for selected projectiles, mostly range from 1 to 7, higher than expected, reflecting complex projectile dynamics. These new perspectives will impact projectile hazard mitigation and the interpretation of projectile deposits from past eruptions, both on Earth and on other planets.

Plain Language Summary Explosive volcanic eruptions launch incandescent fragments, sometimes partially molten, to distances of up to several kilometers from the volcano. The largest fragments, from the size of an apple to that of a van, travel in air following the same laws that control the flight of artillery shells and, on landing, may cause the same harmful consequences. To protect people and properties from these volcanic projectiles, their occurrence in volcanic rocks is documented, and their motion is simulated by computer models. However, both field studies and computer models require validation, but in-flight observation of the projectiles have been sparse, so far. We used state-of-the-art high-speed cameras, filming volcanic projectiles in slow motion to understand and measure the processes that control their flight dynamics. We found that the in-flight deformation, rotation, and collision of the projectiles have a deep impact on their trajectory. We also measured the size of craters left by the projectiles on landing, and we derived specific parameters that are essential to model projectiles flight. We found that currently used models often do not account for all the in-flight dynamics. Our findings will improve interpreting the motion of the projectiles and mitigating their hazard.

1. Introduction

Volcanic ballistic projectiles (VBPs) are centimeter- to meter-sized pyroclasts—i.e., solid to molten rock fragments produced and ejected during explosive volcanic eruptions—that are large enough to move in the atmosphere along ballistic trajectories, mimicking the motion, and often the outcome, of artillery shells. Their very name is suggestive of their harmfulness. Even though in the list of volcano-related casualties they rank below large-scale processes such as pyroclastic density currents (ground-hugging, hot avalanches of gas and pyroclasts), VBPs still represent a constant threat to life and properties in the vicinity of volcanic vents [Blong, 1984; Williams *et al.*, 2017] and are amongst the most frequent causes of fatal accidents on volcanoes [Fitzgerald *et al.*, 2017]. As recently as September 2014, more than 50 people lost their lives to VBPs during an eruption while visiting the summit area of Ontake volcano (Japan) [Oikawa *et al.*, 2016; Tsunematsu *et al.*, 2016]. Indeed, volcano tourists, visiting active volcanoes for their fascination, are particularly at risk from

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VBP's ejected during unexpected or larger-than-usual eruptions (Figure 1). The hazard from VBP's has often prompted the closure of touristic viewpoints and trails at places such as Stromboli volcano (Italy) and Kilauea's Halema'uma'u (Hawaii) and even prompted the development of ad hoc shelters, like those at Stromboli or Sakurajima (Japan) [e.g., Fitzgerald et al., 2017; Dolce et al., 2007]. Volcanologists are perhaps the category of people most vulnerable to VBP's, as in the case of the six colleagues who lost their lives in the January 1993 eruption of Galeras volcano (Colombia) [Baxter and Gresham, 1997]. As exemplified by the Ontake and Galeras cases, often the most harmful VBP's come from small-scale, unexpected eruptions, in contrast with the widespread destruction from larger-scale processes during higher magnitude eruptions.

Like other volcanic products, VBP's hold important information on past eruptions. However, contrary to the case of other products, the physical laws that control the emplacement of VBP's have been the subject of scientific studies for centuries, because of the connatural human instinct for throwing objects and its obvious, crucial applications. From Aristotelian theory of "impetus," or momentum, through Galileo's study of parabolic trajectories, to Euler's analysis of the motion of bodies through a fluid, the governing laws for the motion of projectiles have a long and honorable history. Building on this history, volcanologists have long since mapped the size, shape, and location of VBP's cropping out in volcanic deposits [e.g., Minakami, 1942]. These quantities can be combined to model the possible trajectories followed by projectiles from the vent to their final resting position and eventually reconstruct, or at least estimate, crucial parameters of the driving eruption. The main focus of these reconstructions is, most commonly, on the damage zone and on the ejection velocity of pyroclasts and the related pressure differential at the volcanic vent. However, other important parameters can be derived, including eruptive energy budget, eruption evolution, and vent location and shifts (see below, section 2.1). Theoretical and experimental models have been combined with the field properties of VBP's from ancient eruptions even to infer the density of the Martian atmosphere in the past [Manga et al., 2012].

Closer to us, the size and spatial distributions of VBP's from past eruptions, coupled with ballistic modeling of their trajectory, are key to forecast their possible impact in future eruptions by drawing VBP hazard maps, either focused solely on ballistic projectiles or as an aspect of a multihazard map [Artunduaga and Jimenez, 1997; Alatorre-Ibargüengoitia et al., 2006, 2012; Ferrés et al., 2013; Fitzgerald et al., 2014; Sandri et al., 2014; Konstantinou, 2015; Alatorre-Ibargüengoitia et al., 2016; Biass et al., 2016]. These hazard maps represent an essential component of the hazard mitigation system of any active volcano (together with volcano monitoring systems and specialized communication) in the case of volcanic crises [e.g., Sparks et al., 2013; Fitzgerald et al., 2017]. The reliability of such maps depends largely on both (i) models rooted in the appropriate physical functions and input parameters and (ii) observational validations. In this paper, we first review current geological evidence, theoretical and experimental models, and direct observations concerning VBP's. Then we present the results of the new, high-speed observations of the in-flight behavior of VBP's. These observations serve to document in greater detail the flight dynamics previously described, uncover unexpected dynamics, and finally parameterize properties and processes that were previously precluded from direct observation.

2. State of the Art

2.1. Volcanic Ballistic Projectiles in the Geological Record

The very definition of volcanic ballistic projectiles, i.e., volcanic ejecta that follow ballistic trajectories, is intimately dependent on the highly variable nature of explosive volcanic eruptions. These range 10–10⁴ m in height reached by the eruption products and 1–10⁴ s in ejection duration, function of eruption dynamics [Sigurdsson, 2015]. When driven by the liberation of magmatic gases, eruption style ranges from weak, intermittent Strombolian and Hawaiian, through transient but vigorous Vulcanian, to almost steady state, very vigorous sub-Plinian, Plinian, and Ultraplinian. Eruptions that are not, or only subordinately, driven by magmatic gases are usually transient, and their style is named after the driving expanding phase, such as phreatomagmatic, when driven by the direct interaction of magma with surface or groundwater, and phreatic and hydrothermal, when driven by the expansion of heated water or hydrothermal fluids with no magma being directly involved [Sigurdsson, 2015].

A broad variety of eruption styles and scales results in an equally broad range of processes controlling the motion of the ejecta, and identical pyroclasts leaving the volcanic vent at the same velocity and angle may

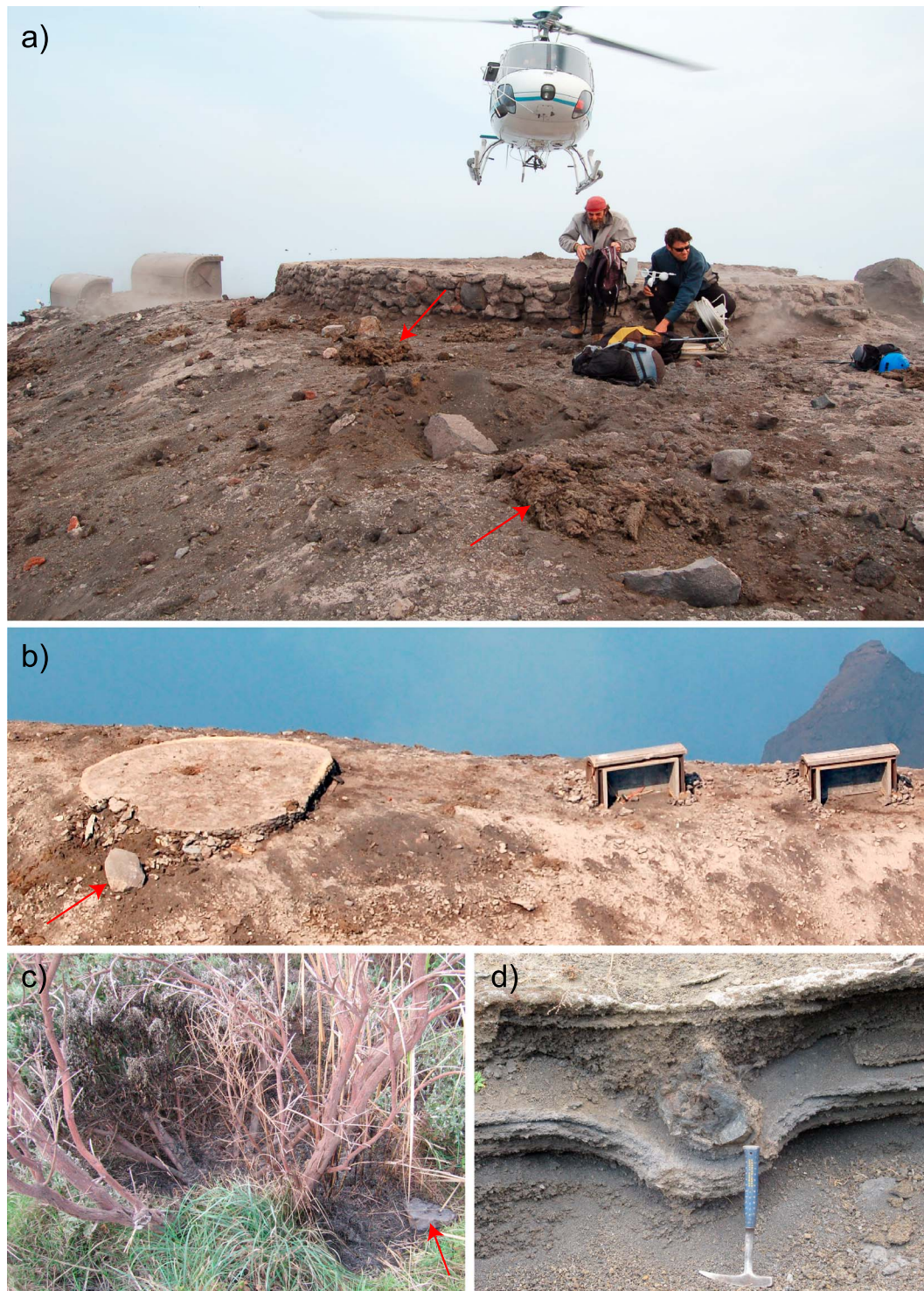


Figure 1. Field pictures of volcanic ballistic projectiles. (a) Projectiles emplaced during the 2007 eruptive crisis at Stromboli volcano (Italy), close to the helipad and along the touristic path that leads to the crater outlook. Molten projectiles deformed on impact with the ground (red arrows), while solid rock ones produced impact craters. (Picture courtesy of Mauro Rosi). (b) Aerial view of the damage caused by a large projectile (red arrow) to the helipad. On the right hand, the shelters installed to protect tourists from projectiles (picture courtesy of the Italian Department of Civil Protection). (c) On impact, projectiles (red arrow) may be hot enough to char vegetation and start wildfires. (d) A volcanic ballistic projectile in the geological record of Xitle volcano (Mexico). On impact with the ground, the projectile deformed the underlying volcanic layers, forming a “bomb sag,” a depression subsequently filled by the products of the same eruption.

follow very different trajectories in different eruptions. For instance, centimeter-sized pyroclasts are often emplaced ballistically in Strombolian eruptions, while being engulfed and convectively lofted in eruption plumes in Plinian eruptions. In Vulcanian eruptions, even meter-sized pyroclasts may follow trajectories that, due to their interaction with the surrounding gas-ash mixture, deviate significantly from the ideal ballistic case [e.g., *de' Michieli Vitturi et al.*, 2010]. For Plinian eruptions, a distinction has been made between ballistic projectiles that are so large that they are less affected by the motions in the eruption column and somewhat smaller ballistic projectiles that are more influenced by the turbulent plume motion [*Self et al.*, 1980].

The first-order question invariably linked to VBPs is how far can they travel? The answer to this question is obviously crucial for the assessment of VBP-related hazard and for planning adequate mitigation actions. Travel range is also the key parameter used to retrieve ejection velocity, eruption intensity, and vent location from the ground distribution of VBPs. Typically, the fallout of VBPs is considered to be a significant hazard within 5 km from the volcanic vent, although 10.4 km is the maximum range of VBPs of far discovered in the geological record (Table 1) [*Alatorre-Ibargüengoitia et al.*, 2012; *Fitzgerald et al.*, 2017]. Eruption style and intensity set the maximum range of VBPs by setting their key ejection parameters, i.e., size, velocity, and angle. However, these well-established control parameters, acquired at ejection, are then joined by other, less-studied and documented in-flight processes. These processes may significantly contribute to set the final range reached by VBPs, as we show in the second part of the paper.

The VBPs are preserved in the geologic record of almost every volcano displaying explosive eruptions (Table 1). In a pyroclastic deposit, the VBPs are usually recognized by being (i) bombs and blocks, i.e., clasts larger than 64 mm with fluidal and blocky shapes, respectively; (ii) outsized with respect to the surrounding clasts; and (iii) inside or nearby structures such as impact craters and bomb sags. Field studies of VBPs usually include their size, density, lithology, impact location, impact angle (from horizontal), orientation (azimuth), and crater size. Generally, the spatial density of VBPs (projectiles per square meter) decreases with increasing distance from the vent [*Kilgour et al.*, 2010], while the shape of impact zones and the size distribution of craters and VBPs varies substantially between eruption styles and magnitudes [*Minakami*, 1942; *Robertson et al.*, 1998; *Pistolesi et al.*, 2008; *Kilgour et al.*, 2010; *Gurioli et al.*, 2013; *Suzuki et al.*, 2013].

Circular VBPs distributions are generally associated with nearly vertical, axis-symmetric eruptions [e.g., *de' Michieli Vitturi et al.*, 2010]. More complex distributions have been observed for laterally directed explosions [e.g., *Alvarado et al.*, 2006], and fan-shape distributions have been also reported, e.g., from a major Strombolian explosion at Stromboli volcano (Italy) [*Gurioli et al.*, 2013, Figure 2] and the 2007 hydrothermal eruption at Ruapehu volcano (New Zealand), where deposits of VBPs follow those of small-scale pyroclastic density currents [*Kilgour et al.*, 2010]. Lateral blasts, such as that of ~1150 B.P. Mount St Helens (USA) [*Mullineaux and Crandell*, 1981] and the 1996 one at Soufrière Hills volcano (Montserrat), generated narrow radial distributions [*Branney and Kokelaar*, 2002] by ejection at low angles [e.g., *Voight*, 1981; *Esposti Ongaro et al.*, 2005] that may even be obstructed by topography to create shadow deposition zones [*Kilgour et al.*, 2010].

In many ash- and block-rich Vulcanian eruption deposits VBPs size tend to increase with distance from the vent, as at Sheveluch (Russia) [*Steinberg*, 1974, 1977], Arenal (Costa Rica) [*Minakami et al.*, 1969; *Fuddali and Melson*, 1971], Ngauruhoe (New Zealand) [*Nairn and Self*, 1978], Ukinrek Maars (USA) [*Self et al.*, 1980], Soufrière Hills [*Druitt and Kokelaar*, 2002], and in the medial to distal (with respect to the eruption vent) deposits at Upper Te Maari (New Zealand) [*Fitzgerald et al.*, 2014]. This distribution can be explained assuming that all VBPs leave the vent with similar velocity and considering that the deceleration due to the drag force is proportional to the surface area/mass ratio, i.e., inversely proportional to VBP diameter (see section 2). Hence, larger VBPs are less affected by drag and more by inertia and therefore can fly to longer distances [e.g., *Minakami*, 1942; *Wilson*, 1972; *McGetchin and Ullrich*, 1973; *Fagents and Wilson*, 1993; *Lorenz*, 2007; *Sottili et al.*, 2012].

The VBPs from phreatomagmatic eruptions, conversely, usually show an overall decrease in diameter with distance from the vent [*Lorenz*, 1970; *Self et al.*, 1980; *Waite et al.*, 1995; *Sottili et al.*, 2012], and the VBPs from the 79 A.D. Plinian eruption of Vesuvius (Italy) follow the same trend [*De Novellis and Luongo*, 2006]. Such a distribution may result from the ejection of VBP in a gas stream [*Lorenz*, 1970], where the smaller projectiles are carried higher into the atmosphere than the larger ones before leaving the eruption column [*De Novellis and Luongo*, 2006].

Table 1. Compilation of Field Observations of Volcanic Ballistic Projectiles in Volcanic Deposits

Eruption	Style	No. of Samples	Maximum Distance (km)	Diameter ^a (m)	Density (kg m ⁻³)	Calculated Velocity (m s ⁻¹)	Model ^b	Reference ^c
2014 Mount Ontake (Japan)	Phreatic		1	0.2	2300 ± 300	145–185	1 (with drag)	1
2014 Mount Ontake (Japan)	Phreatic		1	.1–1	2500	111	2	2
1790 Kilauea (Hawaii)	Phreatic	1721	3.6	0.25–2				9
2012 Upper Te Maari, Tongariro (New Zealand)	Hydrothermal	2215	1.6	CR > 2.5, VPB 0.32	2400	165–310	2	4
2012 Upper Te Maari, Tongariro (New Zealand)	Hydrothermal	3587	2.6	CR 0.3–10.8, VPB 0.36 ± 0.23	2400	200	1 (with drag and gas flow velocity)	5
2007 Mount Ruapehu (New Zealand)	Hydrothermal		1.97	2	1700–2700	135	2	8
Atexcac maar (Mexico)	Phreatomagmatic	43		0.1–2	1400–2900	100–120	2	3
1977 Ukinrek maars (Alaska)	Phreatomagmatic	200	0.7	2–25		100–150	6	19
Sabatini Volcanic District (Italy)	Phreatomagmatic		0.9	0.1–2		50–110	3	7
Big Hole maar (USA)	Phreatomagmatic		3	0.1–2.3	2700		No drag	25
2008 Kilauea (Hawaii)	Strombolian-Hawaiian	696	0.27	1.03	1230	50–80	9	26
2010 Stromboli (Italy)	Strombolian	780	0.43	0.07–4.59	1370–2300	52–70	No drag	10
2003 Stromboli (Italy)	Strombolian-Strombolian	37	2	0.1–1		150	2	11
2007 Stromboli (Italy)	Strombolian-Strombolian	111	1.3	0.3–2	2500	100–210	2	12
2011 Shinmoedake volcano (Japan)	Vulcanian	2	3.4	CR 3.5–4 VPB 0.5–1.1	2100–2400	240–290	2	6
1998–2003 Popocatepetl (Mexico)	Vulcanian	122	3.7	0.2–0.6	2100–2600	180–230	10	27
1999 Guagua Pichincha (Ecuador)	Vulcanian	100	0.8	<1.5	2500	100	2	13
1984–1993 Lascar (Chile)	Vulcanian		5	<2		90–200	3,4	16
1989–1989 Tokachidake volcano, (Japan)	Vulcanian		1	1.8–20		67–96		18
1968 Arenal (Costa Rica)	Vulcanian		5	0.5–1.5	1500	600		21
1935–1941 Asama (Japan)	Vulcanian		4.5	0.9–7.5		130–212	7	22
1975 Ngauruhoe (New Zealand)	Vulcanian		2.8	0.8	2500	400	6	20
1992 Mount Spurr (Alaska)	sub-Plinian	56	3.5	0.1–2	1200–2750	155–840	5	17
1996 Soufriere Hills (Montserrat)	sub-Plinian		2.1	1.2	1100–2100	180	8	23
~ 17,000 B.P. Popocatepetl (Mexico)	Plinian		10.4	0.3–0.4	2100–2700		10	27
79 A.D. Vesuvius (Italy)	Plinian	300	9	0.07–2	600–2700	170–2300		14
1640 B.C. Santorini (Greece)	Plinian	76	7	0.15–1.6				15
1982 Chichón (Mexico)	Plinian		7.1	0.5–0.6			No drag	24

^aCR stands for crater; VPB stands for volcanic ballistic projectile.

^bModel: (1) Tsunematsu et al. [2014]; (2) Mastin [2001]; (3) Fagents and Wilson [1993]; (4) Sherwood [1967]; (5) Waitt et al. [1995]; (6) Wilson [1972]; (7) Minkami [1942]; (8) Bower and Woods [1996]; (9) Biass et al. [2016]; and (10) Alatorre-Ibargüenito and Delgado-Granados [2006].

^cReference: (1) Tsunematsu et al. [2016]; (2) Oikawa et al. [2016]; (3) López-Rojas and Carrasco-Núñez [2015]; (4) Breard et al. [2014]; (5) Fitzgerald et al. [2014]; (6) Maeno et al. [2013]; (7) Sottili et al. [2012]; (8) Kilgour et al. [2010]; (9) Swanson et al. [2012]; (10) Gurio et al. [2012]; (11) Pistoletti et al. [2008]; (12) Pistoletti et al. [2011]; (13) Wright et al. [2007]; (14) De Novellis and Luongo [2006]; (15) Pfeiffer [2001]; (16) Matthews et al. [1997]; (17) Waitt et al. [1995]; (18) Yamagishi and Feebrey [1994]; (19) Self et al. [1980]; (20) Nairn and Self [1978]; (21) Fudali and Nelson [1971]; (22) Minkami [1942]; (23) Robertson et al. [1998]; (24) Yokoyama et al. [1992]; (25) Lorenz [1970]; (26) Houghton et al. [2017]; and (27) Alatorre-Ibargüenito et al. [2012].

In other cases, no obvious trend of VBP and crater size with distance have been observed [Mastin, 1991; Pfeiffer, 2001], suggesting that not all the VBPs are ejected with the same velocity. This is not surprising because the effectiveness of gas-particle coupling in the initial phase of the eruption varies spatially and temporally and is related to the projectile size and density and location within the vent [de' Michieli Vitturi et al., 2010; Breard et al., 2014]. Complex patterns may result from a combination of eruptive mechanisms associated with a dynamic column [Self et al., 1980; Mastin, 1991; De Novellis and Luongo, 2006], VBPs ejected by different vents over overlapping distributions [Breard et al., 2014] and VBP collisions [Tsunematsu et al., 2014; Fitzgerald et al., 2014]. It is noteworthy that the initial velocity of ballistic ejecta is not correlated with eruption scale [Maeno et al., 2013].

The spatial and size distribution of VBPs has been used to infer the morphology and location of eruptive vents for the 1790 eruption of Kilauea volcano (USA) [Swanson et al., 2012], the Baccano maar (Italy) [Buttinelli et al., 2011], and the Atexcac maar (Mexico) [López-Rojas and Carrasco-Núñez, 2015] and to investigate vent development of the Minoan eruption of Santorini volcano (Greece) [Pfeiffer, 2001] and Upper Te Maari crater [Breard et al., 2014].

Many VBPs from the 1992 eruption of Mount Spurr volcano (Alaska) display evidence (e.g., embedding in the downrange side of the impact crater, elongate craters with an asymmetric ejecta rim and ejecta rays radiating on one side) indicating impacting angles distinct to the vertical and an azimuth of the impact angle mostly deviating by 20° to 40° south from the vent azimuth, despite local westerly wind shifting the trajectories eastward [Waite et al., 1995]. Deviations, largely independent of the VBPs diameter, were attributed to the Magnus effect, whereby a particle's trajectory curves in the direction of a sharply applied spin, akin to side spin causing swerve to either side as seen during some baseball pitches [Waite et al., 1995].

Detailed analyses of the ballistic field from observed eruptions are rare. The VBP impact craters from the 2014 eruption of Ontake were mapped 1 day after the eruption from more than 350 airborne images [Kaneko et al., 2016]. From a major explosion of 2010 at Stromboli volcano, Gurioli et al. [2013] measured 780 VBPs both in situ and from hand-held digital photos, recording their long, intermediate and short axis, perimeter, area, weight, and density (Figure 2). The VBPs, featuring ellipsoid shapes and similar thickness, were all flattened on impact and stuck to the impacted ground without breaking, thus preserving their original landing position and shape. A detailed map of the ballistic impact field was obtained for the August 2012 hydrothermal eruption at Tongariro volcano by using more than 300 airborne photographs, orthophotographs, and field measurements [Breard et al., 2014; Fitzgerald et al., 2014]. More than 3587 impact craters from 0.3 to 10.8 m in diameter were identified in the orthophotographs (Figure 3), but field mapping revealed an average ratio of 1 orthophoto-detected crater to 4.5 field-mapped craters, implying a higher concentration of ejecta which are also a relevant source of hazard [Fitzgerald et al., 2014]. All these detailed analyses revealed uneven VBP distributions, with clustering not linked with topographic shielding but with zoning between jets in the plume. Such uneven distributions may limit the use of isopach (deposit thickness) and isopleth (maximum clast size) maps to estimate eruptive volumes for ballistic-dominated eruptions [Gurioli et al., 2013].

Other parameters being equal (e.g., impactor velocity and density, impact angle, substrate characteristics, and other, see section 2.2.5), the size of the impact craters can be correlated with the size of the VBP that generated them. Fudali and Melson [1971] observed an average depth-diameter ratio for 15 impact craters of 1:3.8 at the ballistic field generated by the 1968 eruption of Arenal volcano (Costa Rica). Assuming this ratio, they used empirical relationship between displaced (cratered) mass and projectile kinetic energy for several different target materials to constrain the size, final velocity, and initial velocity from the calculations obtained from a ballistic model. Considering a different approach, two different empirical relationships between projectile diameter (D) and crater size (D_c) have been proposed by Fitzgerald et al. [2014], based on the ballistic impact field generated by the 2012 eruption of Tongariro volcano: a first with $D_c = 0.3507D^{0.3941}$ ($R^2 = 0.51$), considering all substratum lithologies; and a second with $D_c = 0.1178e^{0.3471D}$ ($R^2 = 0.80$), which takes into account only the most common substratum lithology.

Inverse modeling of VBPs distribution have been used to infer eruptive parameters such as ejection velocity and angle (Table 1), representing important constraints to eruption mechanisms. In fact, ejection velocity has been used to estimate the gas content and pressure at the vent considering several eruptive models [e.g., Minakami, 1942; Fudali and Melson, 1971; Wilson, 1980; Fagents and Wilson, 1993; Mastin, 1991; Woods, 1995; Bower and Woods, 1996; Formenti et al., 2003; Taddeucci et al., 2004; Alatorre-Ibargüengoitia et al.,

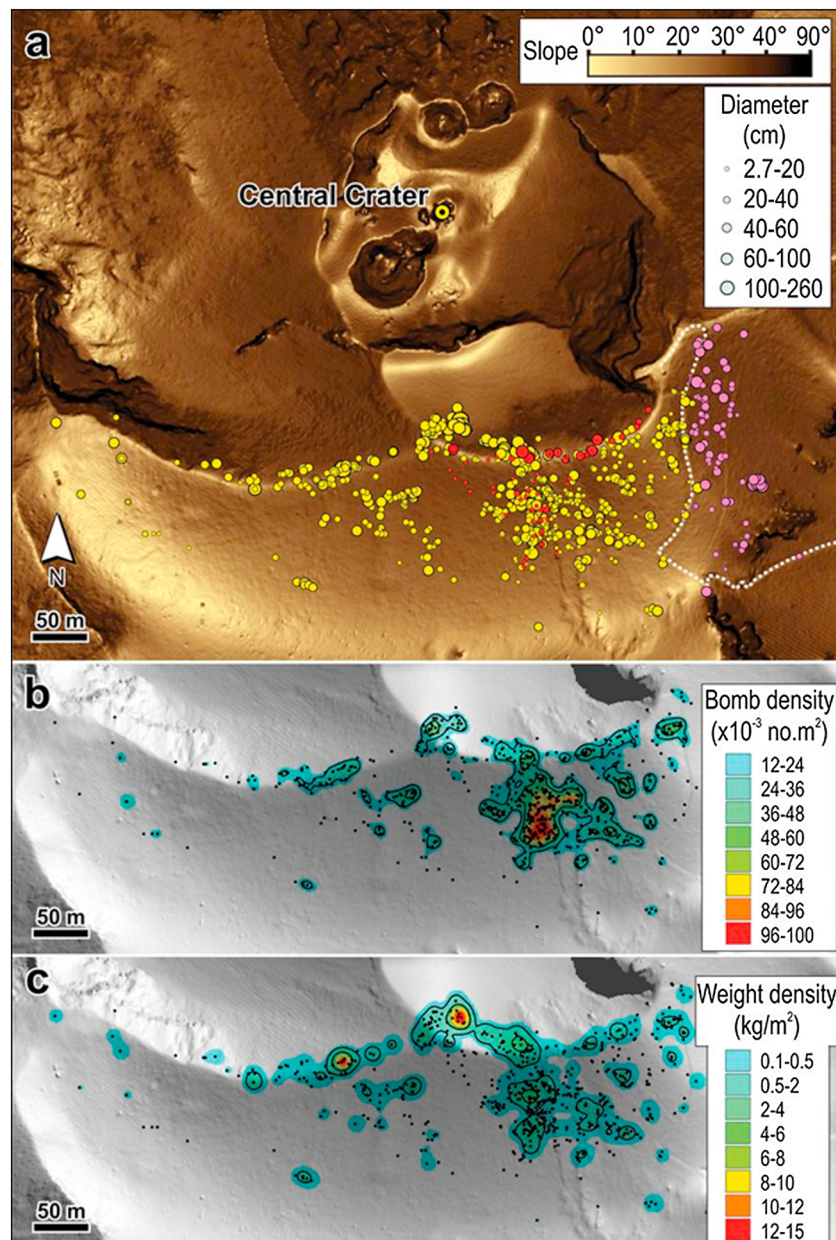


Figure 2. (a) Bomb locations plotted by size (size of circle is function of bomb diameter; purple—24 November 2009; yellow—21 January 2010; red—sampled bombs) over Stromboli crater area slope map. Note the relatively restricted range (few hundreds of meters) of the projectiles. Dashed line is tourist track. (b) Areal bomb densities, in terms of number (no./m^2). (c) Areal bomb densities, in terms of weight. In Figures 2b and 2c, bombs associated with 24 November 2009 event have been excluded, and black dots indicate 21 January 2010 bomb locations. Diameter in key corresponds to average diameter (from Gurioli et al. [2013] reproduced with permission).

2010, 2012; Taddeucci et al., 2012a; Maeno et al., 2013]. Additionally, VBPs trajectories and emplacement have also been used to interpret eruption sequence and dynamics [Wright et al., 2007; Breard et al., 2014; Fitzgerald et al., 2014; Kaneko et al., 2016; Tsunematsu et al., 2016] and to estimate the initial velocity of any simultaneously occurring multiphase flows such as dilute pyroclastic density currents [Fagents and Wilson, 1993; Breard et al., 2014].

Pure ballistic models assuming VBPs to be ejected from a point source into still air may be inadequate [e.g., Fagents and Wilson, 1993; Waitt et al., 1995; De Novellis and Luongo, 2006; de' Michieli Vitturi et al., 2010; Alatorre-Ibargüengoitia et al., 2012; Fitzgerald et al., 2014; Tsunematsu et al., 2016], providing unrealistically

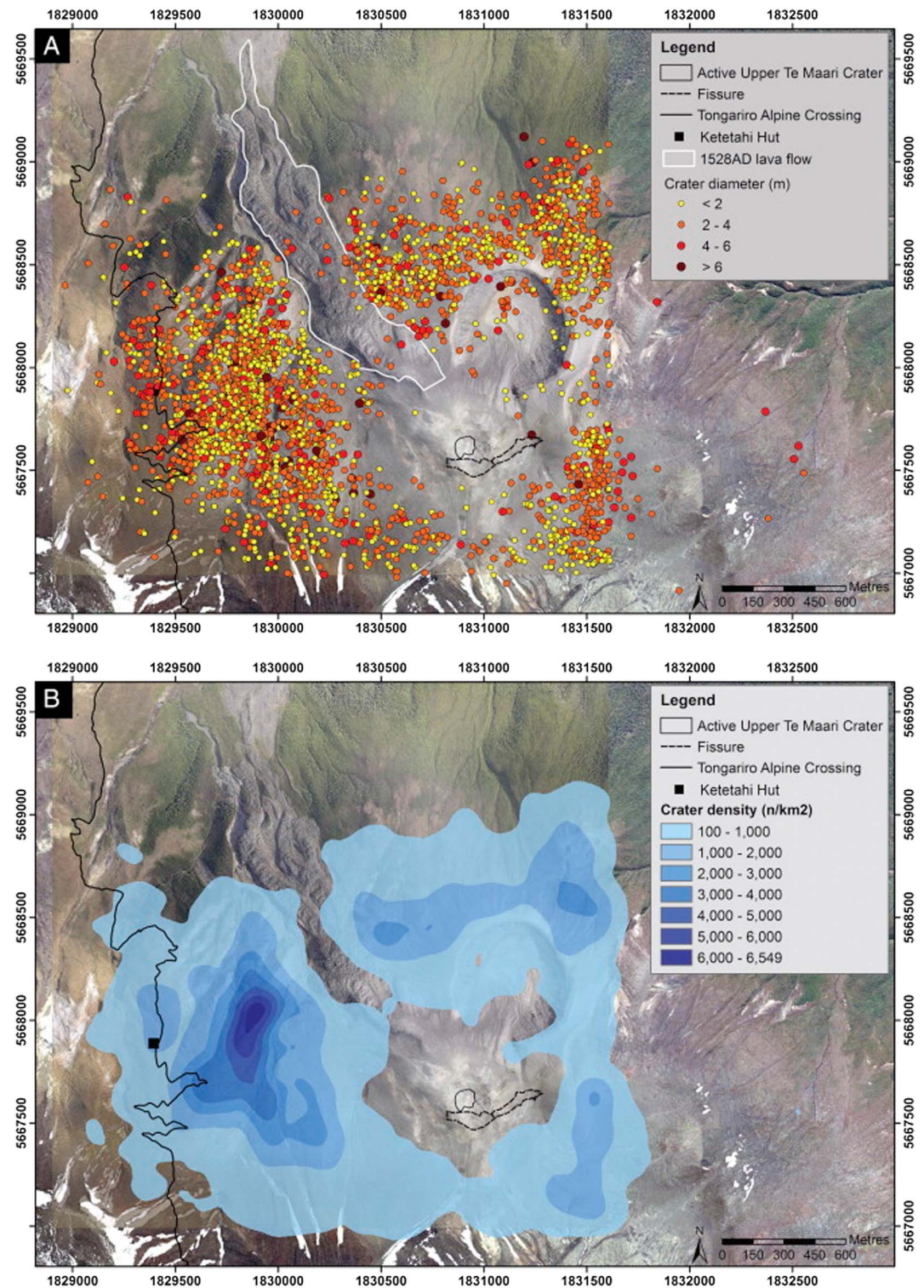


Figure 3. Ballistic impact crater distribution from the August 2012 eruption of Upper Te Maari. Note that projectiles traveling up to 3 km from the source vent. (a) Mean crater diameter is indicated by symbol color. (b) Kernel density of craters per km². (from *Fitzgerald et al.* [2014] reproduced with permission).

high initial velocities, up to 2300 m s^{-1} , which even exceed maximum theoretical values even for an infinite supply of pure gas expanding under frictionless conditions, which is about 2000 m s^{-1} [Mastin, 1995]. In reality, the ejection velocity during explosive eruptions is limited by the speed of sound of the fluid, the presence of solid particles with much higher densities that reduces the ejection velocity of the gas-pyroclasts mixture, and the imperfect coupling between VBPs and the gas during the acceleration phase, which prevents these projectiles to be ejected with the same velocity as the gas. For this reason, the expected ejection velocities of the VBPs are much lower than 2000 m s^{-1} . Furthermore, if these projectiles are assumed to be ejected into a stationary atmosphere, the initial relative velocity of the VBPs with respect to the still air is very high, leading to a large overestimate of the drag force, which escalates considering that the drag force greatly increases as the projectile velocity approaches the speed of sound.

The VBPs are not ejected into still air, and coupling of the gas flow significantly reduces the drag and carries the VBPs further before deposition. This effect is especially important near the vent where the velocities of the VBPs are maximum and it is relevant even for relatively large VBPs [de' Michieli Vitturi et al., 2010]. Current eruption models lack sufficient physical details to explain the complexities of these phenomena, such as the effects of turbulent interactions between the pyroclasts and the gas phase.

2.2. Modeling Ballistic Motion

2.2.1. Forces Acting on Ballistic Projectiles

The equation of motion of a particle in a nonuniform, unsteady, and infinite medium on a rotating planet can be expressed according to Newton's second law as follows:

$$m_p \frac{d\mathbf{v}}{dt} = \mathbf{F}_g + \mathbf{F}_d + \mathbf{F}_{vm} + \mathbf{F}_p + \mathbf{F}_M + \mathbf{F}_l + \mathbf{F}_{bu} + \mathbf{F}_{Ba} + \mathbf{F}_{CE} + \mathbf{F}_{CO} \quad (1)$$

where m_p is the particle mass and $\mathbf{v}$ its velocity (expressed as a vector), t is time, and the total force is expressed in terms of gravity force ($\mathbf{F}_g$), steady state drag force ($\mathbf{F}_d$), virtual mass force ($\mathbf{F}_{vm}$), pressure gradient force ($\mathbf{F}_p$), Magnus force ($\mathbf{F}_M$), lift force ($\mathbf{F}_l$), buoyant force ($\mathbf{F}_{bu}$), Basset force ($\mathbf{F}_{Ba}$), centrifugal force due to the Earth's rotation ($\mathbf{F}_{CE}$) and Coriolis force ($\mathbf{F}_{CO}$). The first six forces (excluding $\mathbf{F}_M$) correspond to the so-called Basset-Boussinesq-Ossen equation, typically termed BBO equation [Maxey and Riley, 1983, and references therein; de' Michieli Vitturi et al., 2010; Bertin, 2017]. Let us present the meaning, expression, and significance of each force with more detail (all the parameters are summarized in Table 2).

Several of these forces, including centrifugal, Coriolis, lift, Basset, and buoyant, have very little influence on ballistic projectiles traveling only a few kilometers. Therefore, these forces can be neglected in most applications, as discussed in the following paragraphs.

The centrifugal force is due to the rotation around the Earth's polar axis, and it is given by

$$\mathbf{F}_{CE} = -m\mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}_p) \quad (2)$$

where $\mathbf{\Omega}$ is the angular velocity vector of the Earth's rotation (with magnitude $7.292 \times 10^{-5} \text{ rad s}^{-1}$) and $\mathbf{r}_p$ is the radius of the parallel expressed as vector (with magnitude $\mathbf{r}_p = R \cos \lambda$ where R is the Earth's radius and λ the latitude). This force is perpendicular to the Earth's rotation axis and always points outward from this axis with a magnitude given by

$$\mathbf{F}_{CE} = m\mathbf{\Omega}^2 R \cos \lambda \quad (3)$$

The Coriolis effect is the apparent deflection of a moving projectile to the right in the Northern Hemisphere and to the left in the Southern Hemisphere, and it is caused by the Coriolis force, which is perpendicular both to the direction of the velocity of the moving projectile and to the Earth's rotation axis according with the following expression:

$$\mathbf{F}_{CO} = -2m\mathbf{\Omega} \times \mathbf{v} \quad (4)$$

The Basset force represents the temporal delay in boundary layer development around the particle as the relative velocity changes with time and it is expressed by [de' Michieli Vitturi et al., 2010]:

$$\mathbf{F}_{Ba} = \frac{3}{2} D^2 \sqrt{\pi \rho \mu} \int_0^t \frac{d(\mathbf{u} - \mathbf{v})}{\sqrt{t - \tau}} d\tau \quad (5)$$

The Basset force accounts for a modified drag on the particle caused by an unsteady motion of a particle in a flow field, where the term $(t - \tau)$ represents the time elapsed since past acceleration from 0 to t [Zhu and Fan, 2016]. This force is expected to increase the drag force when the Reynolds number for relative motions is quite small [de' Michieli Vitturi et al., 2010] and if the particle is accelerated at a high rate, but it can be neglected at small ρ_f over ρ_p ratios [Zhu and Fan, 2016]. Considering that this ratio is usually lower than 10^{-3} in volcanic projectiles, that the variations of the relative velocity of particles are small to the timescales of the fluid flow, and that the projectiles are small compared to the length scale of the fluid, the Basset force has been neglected in ballistic studies [de' Michieli Vitturi et al., 2010; Bertin, 2017].

Buoyant Force, assuming a hydrostatic pressure, is given by

$$\mathbf{F}_{bu} = -\rho_f V_p \frac{\mathbf{g}_0}{\left(1 + \frac{z}{R}\right)^2} \quad (6)$$

Several authors have included the buoyant force [e.g., Mastin, 2001; Bagheri et al., 2013] since it does not add complications to the motion equations. However, since most VBPs have densities higher than 10^3 kg m^{-3} , then for projectiles flying in the Earth's atmosphere $\frac{\mathbf{F}_{bu}}{\mathbf{F}_g} = \frac{\rho_f}{\rho_p} < 10^{-3}$ and therefore buoyant force can be neglected.

Aerodynamic lift force is the force that acts at a right angle to the direction of motion of a projectile through the air due to differences in air pressure, and its magnitude can be expressed as a function of the lift coefficient (C_l) as follows:

$$\mathbf{F}_l = \frac{1}{2} \rho_f C_l A |\mathbf{v} - \mathbf{u}|^2 \quad (7)$$

The lift force is appreciable in strong shear flows where the shear rate and the relative velocity between the particle and the fluid are large [Mei, 1996] and might also be relevant for relatively flat projectiles, even though to date there is no experimental or observational data for volcanic particles. At low shear rates, this force is generally smaller than the drag force and it is believed to be negligible [Bertin, 2017]. Saunderson [2008] found that the maximum range of projectiles increased by up to 3.3% when aerodynamic lift, centrifugal, and Coriolis forces were included in the simulations and therefore considered such forces of second order.

The Magnus effect is the generation of a sideways force on a spinning projectile in a fluid when there is relative motion between the spinning particle and the fluid. It can be calculated as [e.g., Sawicki et al., 2003]

$$\mathbf{F}_M = \frac{1}{2} \rho_f C_l A (\mathbf{v} - \mathbf{u})^2 \frac{\omega \mathbf{n} \times (\mathbf{v} - \mathbf{u})}{|\omega \mathbf{n} \times (\mathbf{v} - \mathbf{u})|} \quad (8)$$

where ω is the angular velocity of spin of the projectile (in radians/second) and $\mathbf{n}$ is a unit vector that is normal to the vertical plane of flight. So far, the Magnus force has not been investigated in ballistic models. Bertin [2017] argued that it can be neglected at very low ρ_a/ρ_p ratios and/or at low angular velocities. However, deposit observations [Waite et al., 1995] suggest that this force may be important if spinning is considerable.

Virtual Mass Force accounts for the inertia added to the projectile due to the acceleration of the portion of the fluid that surrounds it and is given by Bertin [2017]:

$$\mathbf{F}_{vm} = C_{vm} \rho_f V_p \frac{d}{dt} (\mathbf{v} - \mathbf{u}) \quad (9)$$

where C_{vm} is the virtual mass coefficient.

The pressure gradient force ($\mathbf{F}_p$), assuming a constant pressure gradient (∇p) in the proximity of a projectile, can be expressed as [de' Michieli Vitturi et al., 2010]:

$$\mathbf{F}_p = -V_p \nabla p \quad (10)$$

The virtual mass force and pressure gradient force were considered by de' Michieli Vitturi et al. [2010] in a Lagrangian model one-way coupled with the carrier flow field given by a Eulerian multiphase flow code, to

Table 2. Parameter Notation^a

Symbol	Parameter
A	Representative cross-section area of the particle
A_a	Actual surface area
A_s	Surface area of the equivalent sphere
$\mathbf{a} = (a_x, a_z)$	Particle acceleration vector
C_d	Drag coefficient
C_{de}	Effective drag coefficient
C_{dr}	Reduced drag coefficient
C_{ds}	Drag coefficient of a sphere with same volume and Reynolds number
C_l	Lift coefficient
C_R	Restitution coefficient
C_{vm}	Virtual mass coefficient
D	Particle diameter
D_c	Impact crater diameter
D_{eq}	Equivalent diameter of a sphere with the same volume
D_g	Drag acceleration
dT/dz	Temperature rate (in K/km^{-1})
E	Particle elongation
$\mathbf{F}_{Ba}$	Basset force
$\mathbf{F}_{bu}$	Buoyant force
$\mathbf{F}_{CE}$	Centrifugal force due to the Earth's rotation
$\mathbf{F}_{co}$	Coriolis force
$\mathbf{F}_d$	Drag force
$\mathbf{F}_g$	Gravity force
$\mathbf{F}_l$	Lift force
$\mathbf{F}_M$	Magnus force
F_N	Newton's shape description
$\mathbf{F}_p$	Pressure gradient force
F_S	Stoke's shape description
F_{vm}	Virtual mass force
f	Flatness
f_1, f_2	Drag coefficient functions
g	Gravitational acceleration
$\mathbf{g}_o$	Gravitational acceleration vector at sea level
γ	Ratio of specific heats at constant pressure and constant volume (1.4 for air)
l	Smallest dimension measured on the maximum-area projection
k	Nikuradse's sand-grain roughness
k_N	Newton's drag correction
k_S	Stoke's drag correction
L	Largest dimension measured on the maximum-area projection
λ	Latitude
η_a	Dynamic air viscosity
Ma	Mach number
m_p	Particle mass
$\mathbf{n}$	Unit vector that is normal to the vertical plane of flight
p	Fluid pressure
p_g	Initial gas pressure
π_v	Crater efficiency
π_2	Inverse Froude number
θ	Trajectory angle
ρ_a	Air density at sea level
ρ_c	Substratum density
ρ_f	Fluid density
ρ_p	Particle density
R	Earth's radius
r	radial distance
R_a	Gas constant for air ($286.98 \text{ J kg}^{-1} \text{ K}^{-1}$)
r_d	Arbitrary distance for drag reduction
Re	Reynolds number
$Re_{critical}$	Critical Reynolds number
r_o	Distance corresponding to maximum ejection velocity
$\mathbf{r}_p$	Radius of an Earth's parallel expressed as vector

Table 2. (continued)

Symbol	Parameter
S	Smallest dimension measured on the minimum-area projection
S_p	Spin parameter
sgn	Sign of the vertical velocity component
σ	Ratio between cross section area and mass of particle
t	Time
T_o	Temperature at sea level (in Kelvin)
$t - \tau$	Time elapsed since past acceleration from 0 to t
$\mathbf{u} = (u_x, u_z)$	Wind velocity vector
u_g	Gas speed
u_{go}	Initial gas speed
u_r	Radial component of the fluid velocity
$\mathbf{v} = (v_x, v_z)$	Particle velocity vector
v_f	Particle speed just before the impact
V_c	Impact crater volume
V_p	Particle volume
v_r	Radial component of the particle velocity
ω	Angular velocity of spin of the projectile
Ω	Angular velocity vector of the Earth's rotation ($7.292 \times 10^{-5} \text{ rad s}^{-1}$)
x	Horizontal position
Y	Substratum material cohesive strength
ψ	Shape factor
z	Vertical position

^aBold fonts indicate vector quantities.

describes plume dynamics in Vulcanian eruptions. Considering these forces, the movement equations expressed in cylindrical coordinates (x_r, x_z) are

$$\frac{dx_r}{dt} = v_r \quad (11a)$$

$$\frac{dx_z}{dt} = v_z \quad (11b)$$

$$\left(1 + \frac{\rho_f}{2\rho_p}\right) \frac{dv_r}{dt} = \frac{AC_d\rho_a(v_r - u_r)|\mathbf{v} - \mathbf{u}|}{2m_p} - \frac{1}{\rho_p} \frac{dp}{dr} + \frac{\rho_f}{2\rho_p} \frac{du_r}{dt} \quad (12a)$$

$$\left(1 + \frac{\rho_f}{2\rho_p}\right) \frac{dv_z}{dt} = \frac{AC_d\rho_a(v_z - u_z)|\mathbf{v} - \mathbf{u}|}{2m_p} - \frac{1}{\rho_p} \frac{dp}{dz} + \frac{\rho_f}{2\rho_p} \frac{du_z}{dt} - g \quad (12b)$$

where v_r and v_z and u_r and u_z are the radial and vertical components of projectile and carrier flow velocity, respectively. The physical quantities of these equations corresponding to the carrier flow field are obtained from the output variables of the Eulerian multiphase flow code.

Bertin [2017] calculated the virtual mass coefficients for an ellipsoid and evaluated the virtual mass force acting on the VBPs traveling through air and found that this force almost does not contribute to ballistic trajectories and therefore can be considered negligible. Moreover, pressure gradients across projectiles traveling in the atmosphere can be neglected [*de' Michieli Vitturi et al.*, 2010; *Bertin*, 2017].

Drag force corresponds to steady motion at an instantaneous velocity (for a uniform stream velocity) and can be expressed as follows:

$$\mathbf{F}_d = \frac{1}{2} \rho_f C_d A |\mathbf{v} - \mathbf{u}| (\mathbf{v} - \mathbf{u}) \quad (13)$$

where ρ_f is the density of the fluid, C_d is the drag coefficient which depends on the projectile and flow properties (see section 2.2.2), A is the representative cross-section area of the particle, and $\mathbf{v} - \mathbf{u}$ is the relative velocity between the particle and the flow. This is the second most important force acting on the VBPs and has been considered in most ballistic studies.

Finally, gravity force is the main force acting on ballistic projectiles. On a homogeneous sphere this force depends on altitude according to the following expression:

$$\mathbf{F}_g = m_p \frac{\mathbf{g}_0}{\left(1 + \frac{z}{R}\right)^2} = \rho_p V_p \frac{\mathbf{g}_0}{\left(1 + \frac{z}{R}\right)^2} \quad (14)$$

where ρ_p and V_p are the particle density and volume, respectively, $\mathbf{g}_0$ is the gravitational acceleration at sea level, and z is the altitude. Generally, it is assumed that the gravity force is constant, but the corresponding value has to be taken into account in other planetary bodies [e.g., Kerber *et al.*, 2011].

Most studies have considered solely gravity and drag forces, because they are thought to be the most important ones [e.g., Minakami, 1942; Sherwood, 1967; Wilson, 1972; Self *et al.*, 1980; Waitt *et al.*, 1995; Alatorre-Ibargüengoitia *et al.*, 2006]. In this case, the motion equation can be expressed in a two-dimensional rectangular coordinate system as follows [e.g., Wilson, 1972; Waitt *et al.*, 1995]:

$$\frac{dx}{dt} = v_x \quad (15a)$$

$$\frac{dz}{dt} = v_z \quad (15b)$$

$$\frac{a_x = dv_x}{dt} = - \frac{AC_d \rho_a(z)(v_x - u_x)|\mathbf{v} - \mathbf{u}|}{2m_p} \quad (16a)$$

$$\frac{a_z = dv_z}{dt} = - \frac{AC_d \rho_a(z)(v_z - u_z)|\mathbf{v} - \mathbf{u}|}{2m_p} - g \quad (16b)$$

where x and z are the horizontal and vertical position coordinates, respectively, $\mathbf{v} = (v_x, v_z)$ and $\mathbf{a} = (a_x, a_z)$ are the velocity and acceleration vectors of the projectile, respectively, t is the time, A and m_p are the frontal area and mass of the VBP, respectively, C_d is the drag coefficient, $\rho_a(z)$ is the air density as a function of altitude, $\mathbf{u} = (u_x, u_z)$ is the wind velocity (most models assumed $u_z = 0$ throughout the trajectory), $|\mathbf{v} - \mathbf{u}| = \sqrt{(v_x - u_x)^2 + (v_z - u_z)^2}$, and g is the gravitational acceleration. Considering variable drag coefficient and air density, and without further simplifications, (equations (15a), (15b), (16a) and (16b)) can be integrated throughout the trajectory using a fourth-order Runge-Kutta method [Wilson, 1972] which is simple, stable, self-starting, and with accuracy sufficient for this kind of calculations. The initial position is $(0, z_0)$, where z_0 is the altitude of the vent above sea level, and the initial velocity is $(v_0 \cos \theta_0, v_0 \sin \theta_0)$, where θ_0 is the ejection angle with respect to the horizontal axis. Trajectory calculation is made during discrete time intervals until $z = z_f$, i.e., the altitude of the landing point.

Equations (15a), (15b), (16a) and (16b) can also be solved analytically assuming that the drag coefficient and air density are constant throughout the trajectory [Minakami, 1942; Sherwood, 1967; Self *et al.*, 1980]. In order to uncouple these equations, the product $(v_x - u_x)|\mathbf{v} - \mathbf{u}|$ in equation (16a) must be replaced with $(v_x - u_x)^2 \text{sgn}(v_x - u_x)$ and $v_z|\mathbf{v} - \mathbf{u}|$ in equation (16b) must be replaced with $v_z^2 \text{sgn } v_z$ where “sgn” refers to the sign of these terms. These simplifications result in underestimating the drag force at each point along the trajectory by the factor $\sqrt{(\sin^4 \theta + \cos^4 \theta)}$ where θ is the trajectory angle [Sherwood, 1967]. Integrating twice (equations (16a) and (16b)) considering these assumptions gives the position of the trajectory of the projectile as a function of time:

$$x = \frac{2}{C_d \rho_a \sigma} \ln \left[\frac{1}{2} C_d \rho_a \sigma (v_0 \cos \theta_0 - u) t + 1 \right] + ut \quad (17a)$$

$$z = \begin{cases} \frac{2}{C_d \rho_a \sigma} \ln \left[\frac{1}{2} C_d \rho_a \sigma v_0 \sin \theta_0 t + 1 \right] - \frac{1}{2} g t^2 & v_z \geq 0 \\ \frac{-2}{C_d \rho_a \sigma} \ln \left[\cosh \left((t - t_0) \sqrt{\frac{1}{2} C_d \rho_a \sigma g} \right) \right] - \frac{1}{2} g t^2 & v_z < 0 \end{cases} \quad (17b)$$

where $\sigma = \frac{A}{mp}$. Only a few ballistic studies have considered solely gravity force [e.g., Yokoyama *et al.*, 1992] to use a simple analytical solution or to investigate the influence of other effects such as collisions [Tsunematsu *et al.*, 2014].

2.2.2. Parameter Determination

The VBPs density refers to their bulk density which can be either measured [e.g., Waitt *et al.*, 1995; Alatorre-Ibargüengoitia and Delgado-Granados, 2006; Alatorre-Ibargüengoitia *et al.*, 2012; Gurioli *et al.*, 2013; Fitzgerald *et al.*, 2014; Tsunematsu *et al.*, 2016] or assumed [e.g., Chouet *et al.*, 1974; Patrick *et al.*, 2007; de' Michieli Vitturi *et al.*, 2010; Kilgour *et al.*, 2010; Vanderkluysen *et al.*, 2012; Maeno *et al.*, 2013; Tsunematsu *et al.*, 2014; Konstantinou, 2015]. Projectile densities mostly fall in the range from 950 to 2700 kg m⁻³ (Table 1) [Waitt *et al.*, 1995; Tsunematsu *et al.*, 2014].

The surface area/mass ratio (σ) depends on the shape and orientation of the VBP and can be quite diverse [e.g., Waitt *et al.*, 1995; Alatorre-Ibargüengoitia and Delgado-Granados, 2006]. For simplicity, most ballistic models have calculated this ratio considering the equation corresponding to spheres or equivalent to spheres [de' Michieli Vitturi *et al.*, 2010; Alatorre-Ibargüengoitia *et al.*, 2012; Vanderkluysen *et al.*, 2012; Gurioli *et al.*, 2013; Tsunematsu *et al.*, 2014; Konstantinou, 2015; Biass *et al.*, 2016]:

$$\sigma = \frac{A}{m_p} = \frac{3}{2\rho_b D_{eq}} \quad (18)$$

where D_{eq} is the equivalent diameter, usually taken as the geometrical average of two or three perpendicular diameters [e.g., Alatorre-Ibargüengoitia and Delgado-Granados, 2006]. Bertin [2017] calculated the projected A_a corresponding to ellipsoids with arbitrary orientation without rotation as a function of three different semiaxis lengths and particle's inclination with respect to the horizontal plane and its azimuth at a given time.

Air density (ρ_a) can be calculated using altitude empirical equations fitting air density data [Waitt *et al.*, 1995; Alatorre-Ibargüengoitia and Delgado-Granados, 2006] or can be calculated as a function of height considering air to be a perfect gas, according to the equation [e.g., Bertin, 2017]:

$$\rho_a(z) = \rho_{ao} \left(1 - \frac{z}{T_o} \frac{dT}{dz} \right)^{\frac{g}{R_a}} \quad (19)$$

where ρ_a and T_o are the air density and temperature (in Kelvin) at sea level, respectively, R_a is the gas constant for air (286.98 J kg⁻¹ K⁻¹) and dT/dz is the temperature lapse rate (in K/km⁻¹). These variables characterized the specific atmospheric conditions at a particular volcano.

Experimental data show that the lift coefficient C_l required for lift and Magnus forces is only a weak function of Reynolds number but strongly depends on the spin parameter (S_p):

$$S_p = \frac{D\omega}{2|\mathbf{v} - \mathbf{u}|} \quad (20)$$

To date, there is no lift coefficient data for volcanic particles but there is a bilinear empirical fit for experimental data on rotating baseballs [Sawicki *et al.*, 2003]:

$$\begin{aligned} C_l &= 1.5S_p, & S < 0.1, \\ C_l &= 0.09 + 0.6S_p, & S > 0.1 \end{aligned} \quad (21)$$

Drag coefficient has been considered constant by several VBP studies, mainly for simplicity: the most common used value is $C_d = 1$ [e.g., Sherwood, 1967; McGetchin and Ullrich, 1973; Self *et al.*, 1980; Bower and Woods, 1996; Breard *et al.*, 2014], but other values have been considered such as $C_d = 0.1$ [Kaneko *et al.*, 2016], $C_d = 0.3$ [López-Rojas and Carrasco-Núñez, 2015], $C_d = 0.7$ [Brož *et al.*, 2014; Fitzgerald *et al.*, 2014], $C_d = 0.8$ [Fudali and Melson, 1971; Tsunematsu *et al.*, 2016], and $C_d = 0.65$ for subsonic flow and $C_d = 1.25$ for supersonic flow [Steinberg and Babenko, 1978; Steinberg and Lorenz, 1983]. However, experimental results [e.g., Hoerner, 1965] indicate that C_d in general depends on Reynolds number (Re) and Mach number (Ma). Re for external flow relates the relative importance of viscous versus inertial force, and it is expressed as

$$Re = vD\rho_a/\eta_a \quad (22)$$

where η_a is the dynamic gas viscosity that for air can be calculated from the empirical relation as [List, 1984; Jacobson, 1999; Bertin, 2017]:

$$\eta_a = 1.8325 \times 10^{-5} \left(\frac{416.16}{T_o - \frac{dT}{dz} + 120} \right) \left(\frac{T_o - \frac{dT}{dz}}{296.16} \right)^{1.5} \quad (23)$$

Ma reflects the effects on the flow due to compression of the fluid and can be calculated using the relation for sound speed in a perfect gas as

$$Ma = \frac{v}{\sqrt{\gamma R_a (T_o - \frac{dT}{dz})}} \quad (24)$$

where γ is the ratio of specific heats at constant pressure and constant volume ($\gamma = 1.4$ for air). If the Mach number of a VBP is higher than about 0.5 ($\sim 175 \text{ m s}^{-1}$ for air at $T = 25^\circ\text{C}$ at sea level), C_d is not only a function of Re but also depends on Ma because the drag force increases as pressure waves accumulate ahead of the particle [e.g., Mastin, 2001]. At $Ma = 1-2$, pressure waves coalesce to a single shock wave and C_d reaches a maximum.

Walker et al. [1971] observed the terminal velocities from free fall experiments on pyroclasts with sizes from 0.125 to 16 mm to match well with those theoretically predicted for cylinders of the same size range. Following these results, C_d values corresponding to cylinders as a function of Re has been widely used [Wilson, 1972; Self et al., 1974; Nairn and Self, 1978; Fagents and Wilson, 1993], together with the correction factors for Ma up to 5, after Hoerner [1965]. Waitt et al. [1995] claimed that the initial velocities for VBPs calculated with those data were too high for the expected eruptive pressure conditions and used C_d values from spheres in order to decrease the calculated ejection velocities.

Mastin [2001] considered C_d values only as a function of Re for spheres, cubes (frontal and with the apex facing the flow) when $Ma < 0.5$ and, at higher Ma , corrected C_d values as a function of Ma after Hoerner [1965]. Similarly, Bertin [2017] associated a unique C_d value to each pair of (Re , Ma) as follows:

$$Ma < 0.5 \Rightarrow C_d = \begin{cases} f_1(Re, Ma) \Leftrightarrow 200 \leq Re \leq 2 \times 10^5 \\ f_2(Re, Ma) \Leftrightarrow 2 \times 10^5 < Re \end{cases} \quad (25a)$$

$$0.5 \leq Ma < 1 \Rightarrow C_d = \begin{cases} f_1(Re, Ma) \Leftrightarrow 200 \leq Re \leq 2 \times 10^5 \\ f_2(Re, Ma) \Leftrightarrow 2 \times 10^5 < Re \end{cases} \quad (25b)$$

$$\begin{aligned} 1 \leq Ma < 1.5 &\Rightarrow C_d = 0.95 \\ 1.5 \leq Ma < 2.5 &\Rightarrow C_d = 0.98 \\ 2.5 \leq Ma &\Rightarrow C_d = 0.93 \end{aligned} \quad (25c)$$

where f_1 and f_2 are different drag coefficient functions fitting data from Hoerner [1965] for Re below and above the critical $Re = 2 \times 10^5$, where an interpolation routine can be used [Mastin, 2001]. This procedure only works for spheres. For other shapes, Bertin [2017] proposed the “effective drag coefficient” (C_{de}) adjusted as a fourth-order polynomial function of the shape factor (Ψ) of the data presented by Crowe et al. [2012] as follows:

$$C_{de} = C_d \left[1 + 11.858(1 - \Psi) + 29.356(1 - \Psi)^2 - 72.893(1 - \Psi)^3 + 79.181(1 - \Psi)^4 \right] \quad (26)$$

where Ψ is defined in terms of the surface area of the equivalent sphere (A_s) and the actual surface area (A_a) [Wadell, 1933; Bertin, 2017]:

$$\Psi = \frac{A_s}{A_a} \quad (27)$$

For a given particle of volume V , A_s can be calculated as [Wadell, 1933; Bertin, 2017]

$$A_s = \pi^{1/3} (6V)^{2/3} \quad (28)$$

Bertin [2017] calculated A_a corresponding to ellipsoids with arbitrary orientation. According to his simulations, the launch velocity and the ejection angle of the projectile are first-order parameters influencing

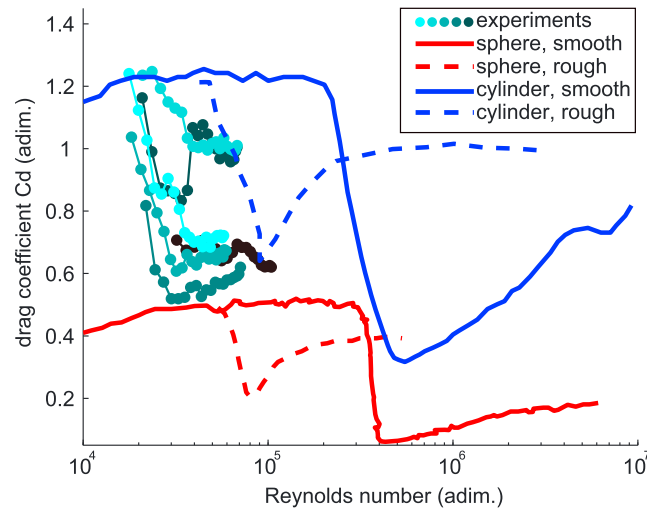


Figure 4. Experimental drag coefficient data at $Re > 10^4$ for volcanic particles (experiments, data from *Alatorre-Ibargüengoitia and Delgado-Granados* [2006]), spheres, and cylinders, both smooth and with a roughness coefficient k/D (where k is Nikuradse's sand grain roughness) of 17.3×10^{-3} and 4.5×10^{-3} , respectively (data from *Achenbach* [1971, 1972] and *Sabersky et al.* [1989]). At a certain $Re_{critical}$ (which decreases with decreasing k/D) the drag coefficient drastically diminishes.

ballistic trajectories. The second-order parameters are the VBP density and the minor radius of the projectile, whereas the intermediate radius and the major radius of the projectile are of third order. His results also highlight the importance of a variable shape-dependent drag and front area determination that cannot be neglected in any ballistic model.

Alatorre-Ibargüengoitia and Delgado-Granados [2006] measured experimentally the drag coefficient for six volcanic pyroclasts from Popocatepetl volcano (Mexico) in a subsonic wind tunnel (Figure 4). They observed that the C_d decreases abruptly at $Re \sim 2 \times 10^4$. This abrupt drop is explained with the transition from a laminar boundary layer to a turbulent one as observed for spheres and cylinders [e.g., *Hoerner*, 1965]. The Re value where this change occurs is defined as $Re_{critical}$. Experimental data show that $Re_{critical}$ decreases for objects with rough surfaces in comparison with the same body shapes but with smooth surfaces (Figure 4), and for this very effect a golf ball “roughened” by inverse topography (dimples) travels farther than a smooth one [*Jorgensen*, 1993; *Crowe et al.*, 2012]. The observed $Re_{critical}$ for volcanic pyroclasts by *Alatorre-Ibargüengoitia and Delgado-Granados* [2006] ($\sim 2 \times 10^4$) is lower than that for spheres and cylinders, possible due to the rough texture of the VBP. The C_d values measured by *Alatorre-Ibargüengoitia and Delgado-Granados* [2006] at $Re > Re_{critical}$ range from 0.62 to 1.01 depending on the sample. *de' Michieli Vitturi et al.* [2010] observed that $C_d \geq 1$ allowed them to semiquantitatively reproduce the modeled trend of dispersal of large pyroclasts for the August 1997 Vulcanian explosions of Soufriere Hills volcano (Montserrat), where meter-sized projectiles traveled farther than decimeter-sized ones. Lower C_d values failed to reproduce this trend with their model.

Bagheri and Bonadonna [2016] measured experimentally the drag coefficient for 300 regular and irregular particles for subcritical particle Reynolds number (i.e., $Re < 3 \times 10^5$) including volcanic particles from different volcanoes. The experiments were carried out in settling columns in a 4 m high vertical wind tunnel developed by *Bagheri et al.* [2013] at $1000 \leq Re < 3 \times 10^5$ (i.e., Newton's regime). Additionally, they considered 881 experimental points from published experimental studies for particles of regular shapes falling in liquids. Combining all the data, they proposed a new C_d versus Re correlation based on two new shape descriptors first proposed by *Ganser* [1993]: Stoke's drag correction (k_s) and Newton's drag correction (k_N) defined as

$$k_s = \frac{C_d}{C_{d,s}} = \frac{C_d}{24/Re} \quad (29)$$

$$k_N = \frac{C_d}{C_{d,s}} = \frac{C_d}{0.462} \quad (30)$$

Table 3. Newton's Drag Correction (k_N) and Drag Coefficient (C_d) for a Number of Newton's Shape Descriptor F_N Calculated Using (Equations (34) and (36))

F_N	k_N	C_d
0.001	21.64	10.39
0.005	10.64	5.11
0.010	7.83	3.76
0.050	3.84	1.84
0.100	2.82	1.35
0.500	1.37	0.66
1.000	1.00	0.48

where $C_{d,s}$ is the drag coefficient of a sphere with same volume and Reynolds number as the particle. As the particle shape tends to a sphere, both correction factors approach unity. k_S represents a correction factor for the Stokes' regime ($Re < 0.1$) and k_N the correction factor for the Newton's regime ($1000 \leq Re < 3 \times 10^5$). After performing detailed and precise size characterization including a number of shape descriptors of all the particles, they

found that k_S is almost equally sensitive to both elongation and flatness, with slightly higher sensitivity to elongation, whereas k_N is more sensitive to flatness than to elongation. Accordingly, they defined the Stoke's and Newton's shape descriptors F_S and F_N as follows:

$$F_S = f e^{1.3} \left(\frac{D_{eq}^3}{LIS} \right) = \frac{D_{eq}^3}{L^{2.3} l^{0.7}} \quad (31)$$

$$F_N = f^2 e \left(\frac{D_{eq}^3}{LIS} \right) = \frac{S D_{eq}^3}{L^2 l^2} \quad (32)$$

where L and l are defined as the largest and smallest dimensions measured on the maximum-area projection, S is the smallest dimension measured in the minimum-area projection, D_{eq} is the diameter of a sphere with the same volume, and $f = S/l$ and $e = l/L$ are the flatness and elongation, respectively. According to their wind tunnel measurements, they obtained the following empirical fits between k_S and F_S and k_N and F_N for $\rho_f/\rho_f > 150$:

$$k_S = \frac{1}{2} \left(F_S^{a_1} + \frac{1}{F_S^{b_1}} \right) \quad (33)$$

$$\log(k_N) = 0.45 [-\log(F_N)]^{0.99} \quad (34)$$

where $0.05 < a_1 < 0.55$ and $0.29 < b_1 < 0.35$. The upper extreme curve for k_S occurs for $a_1 = 0.05$ and $b_1 = 0.35$. These equations were calibrated experimentally for different particle shape and k_S and k_N values up to 15. *Bagheri and Bonadonna* [2016] observed that all data points obtained for freely falling particles irrespective of their size for any subcritical Reynolds number follow a similar trend that can be expressed according to the drag corrections k_S and k_N :

$$\frac{C_d}{k_N} = \frac{24k_S}{Re k_N} \left(1 + 0.125 (Re k_N / k_S)^{2/3} \right) + \frac{0.46}{1 + 5330 / (Re k_N / k_S)} \quad (35)$$

For $1 \times 10^4 \leq Re < 3 \times 10^5$ and assuming $k_N > k_S$ this equation can be approximated by

$$C_d = 0.48 k_N \quad (36)$$

Drag coefficient calculated for several F_N values using equations (34) and (36) are presented in Table 3, whereas Figure 5 shows the impact of flatness and elongation on k_N according to their experimental measurements in the vertical wind tunnel. Interestingly, *Bagheri and Bonadonna* [2016] observed that the effect of surface roughness and vesicularity on the drag coefficient of irregular particles at $8 \times 10^3 \leq Re < 6 \times 10^4$ is only about 10%, which is negligible compared to the effect of particle shape, such as F_N . Furthermore, no correlation was found between k_N and the density ratio when this ratio is higher than 100.

2.2.3. Drag Reduction Near the Vent

Drag reduction near the vent was first introduced quantitatively by *Fagents and Wilson* [1993]. They assumed that erupted pyroclasts accelerate to a maximum speed equal to that of the moving gas (u_{go}) at some distance (r_o) and time (t_o) from their initial position. After t_o , the speed of the moving gas decays with the radial distance (r) at time (t) according to

$$u_g = u_{go} \left(\frac{r_o}{r} \right)^2 e^{-t/\tau} \quad (37)$$

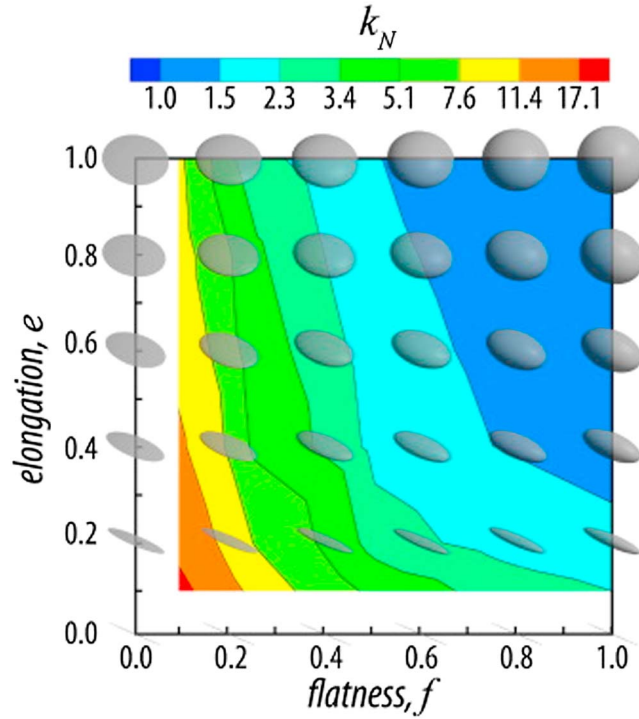


Figure 5. Impact of flatness f and elongation e on the Newton's drag coefficient k_N according to the measurements in the vertical wind tunnel by Bagheri and Bonadonna [2016] (reproduced with permission).

where the time constant τ is assumed to depend on the ratio of initial gas pressure (P_g) to atmospheric pressure (P_a)

$$\tau = \frac{P_g}{P_a} t_o \quad (38)$$

Mastin [2001] considers an arbitrary distance (r_d) from the vent over which there is a reduced drag coefficient (C_{dr}) according to the following expression:

$$C_{dr} = C_d \left(\frac{r}{r_d} \right)^2 \quad (39)$$

It is believed that the drag reduced zone is in the order of tens to hundreds of meters [Mastin, 2001].

The coupled Lagrangian-Eulerian model by de' Michieli Vitturi *et al.* [2010] can be used to calculate the drag force that the eruptive plume exerts on VBPs, including their acceleration and deceleration phases. Instead of considering initial ejection velocity and launching angle of the particles, it is assumed that these projectiles are initially at rest at some point within the conduit and are accelerated by the multiphase expanding

flow. This model accounts for the complex dynamics of the particles that cannot be mimicked by a simple reduced drag region close to the vent as assumed by Fagents and Wilson [1993] and Mastin [2001]. They found that the carrier flow plays a fundamental role even for meter-sized projectiles: for instance, including the effect of the carrier flow will increase the maximum range of a 1 m projectile by about 70%, assuming the same initial velocity. They also observed that the initial depth of the particle in the conduit influences mainly the ejection velocity while the radial position with respect to the conduit axes controls mainly the distance reached by the projectile. It is therefore likely that the near-vent geometry can significantly control the distribution of the projectiles.

2.2.4. Modeling of VBP Collisions

A numerical model considering in-flight collisions of thousands of projectiles was first developed by Tsunematsu *et al.* [2014] and later improved by Tsunematsu *et al.* [2016] by incorporating the drag force. This model simulates an eruption as a series of successive bursts, defined as the simultaneous ejection of a number of spherical particles with a certain size and density. Each ejected particle is characterized by its three-dimensional offset position with respect to the crater vent and velocity defined by its magnitude and its direction given by three angles. According to this model, a collision occurs when the distance between two particles is smaller than the sum of their respective radius. If $\mathbf{v}_1$ and $\mathbf{v}_2$ are the precollision velocities of the colliding particles, it can be shown from conservation of momentum that the postcollision velocities $\mathbf{v}'_1$ and $\mathbf{v}'_2$ can be estimated from [Tsunematsu *et al.*, 2014]:

$$\begin{aligned} \mathbf{v}'_1 &= V'_1 \hat{\mathbf{e}} + \mathbf{v}_1 - (\mathbf{v}_1 \cdot \hat{\mathbf{e}}) \hat{\mathbf{e}} \\ \mathbf{v}'_2 &= V'_2 \hat{\mathbf{e}} + \mathbf{v}_2 - (\mathbf{v}_2 \cdot \hat{\mathbf{e}}) \hat{\mathbf{e}} \end{aligned} \quad (40)$$

with

$$\begin{aligned} V'_1 &= \frac{(1 + C_R)(-\mathbf{v}_1 \cdot \hat{\mathbf{e}} + \mathbf{v}_2 \cdot \hat{\mathbf{e}})}{m_1/m_2 + 1} + \mathbf{v}_1 \cdot \hat{\mathbf{e}} \\ V'_2 &= \frac{(1 + C_R)(-\mathbf{v}_2 \cdot \hat{\mathbf{e}} + \mathbf{v}_1 \cdot \hat{\mathbf{e}})}{m_2/m_1 + 1} + \mathbf{v}_2 \cdot \hat{\mathbf{e}} \end{aligned} \quad (41)$$

where $\hat{e}$ is the unit vector connecting the center of the two particles and C_R is the restitution coefficient which accounts for the fraction of energy lost in the collision. The results show that the VBPs range increases if the net transfer of momentum due to collisions is upward and away from the vent. This transfer is optimum for the case of a limited number of collisions where the mass of the target projectile is smaller than that of the collider, while too many collisions per particle tend to have a neutralizing effect. The distribution of VBPs on the ground is clearly affected by the occurrence of collisions, a point that has to be taken into account when retrieving source parameters such as ejection velocity from field observations. The most efficient collisions (i.e., where the minimum kinetic energy is lost) occur between particles ejected in different bursts, and therefore, this process is particularly important for pulsating explosive volcanic eruptions such as Strombolian, violent Strombolian, and Vulcanian eruptions [Tsunematsu *et al.*, 2014].

2.2.5. Crater Size and Projectile Size

Dimensional analysis shows that crater formation can be expressed in terms of four-dimensionless groups [Holsapple, 1993]: (1) the crater efficiency (π_v) defined as the ratio of the mass of the crater ejecta to that of the impactor; (2) the inverse Froude number (π_2) which is the ratio of the lithostatic pressure at a characteristic depth equal to one projectile radius ($\rho_p g D/2$) to the dynamic pressure generated by the impactor ($\rho_p v_f^2/2$) where v_f is the particle speed (i.e., the absolute value of the velocity vector) just before the impact; (3) the ratio of a substratum material cohesive strength (Y) to the initial dynamic pressure, denoted by π_3 ; and (4) the ratio of the mass densities between the substratum density (ρ_c) and impactor density (ρ_p). The first three terms can be expressed as follows:

$$\pi_v = \frac{\rho_c V_c}{m_p} \quad (42)$$

$$\pi_2 = \frac{gD}{v_f^2} \quad (43)$$

$$\pi_3 = \frac{Y}{\rho_c v_f^2} \quad (44)$$

where V_c is the impact crater volume. If the strength of the soil surface is large compared to the lithostatic pressure, the latter can be ignored. This case corresponds to the strength regime, where any linear dimension of the crater increases linearly with the projectile diameter. On the other extreme, when the substratum strength is small compared to the lithostatic pressure term, the former can be ignored. This is the gravity regime, where the crater volume is not proportional to the impactor volume or mass, nor is it necessarily proportional to its kinetic energy [Holsapple, 1993]. In any case, given the complexity of the impact process and the number of variables, there is no general theoretical relationships between the dimensionless parameters; therefore, they must be determined by experiments or observations.

2.2.6. Summary of the Model Section

While considerable progress has been made to model the trajectory and range of VBPs, there are still first-order challenges that cannot be solved solely with theoretical and numerical models, including (1) the influence of the dynamics of the eruptive plume and/or expanding gas on the ejection of VBPs and resulting drag reduction near the vent; (2) the drag coefficient values measured in situ for VBPs; (3) the effect of rotation of VBPs on the effective area and the possible influence of the Magnus effect; and (4) the in-flight changes of the VBP size including elongation of ductile bombs and in-flight breaking. Fundamental information for these challenges can be obtained from high-speed and high-resolution direct observations of VBPs ejected during different types of volcanic eruptions.

2.3. In-Flight Observation and Measurement of Volcanic Ballistic Projectiles

In comparison with geological observations and modeling, direct observation of VBPs in flight and during emplacement are rare, despite dating as far back as the first direct witness of volcanic eruptions. An interesting compilation on the topic of early observations is found in Mercalli [1907], who, for instance, reports: (i) Sir W. Hamilton observing projectiles at Vesuvius killing two and burning houses at Ottajano village (nowadays called Ottaviano), more than 5 km away from the vent, on 8 August 1779; (ii) a projectile with volume 45 m³ being erupted at Vulcano in the 1889–1890 eruptive crisis; and (iii) L. Spallanzani describing the trajectory of projectiles at Stromboli in A.D. 1788.

In the past century, advancement in photography led to the application of photoballistic techniques to volcanic projectiles, especially at volcanoes characterized by the relatively weak (hence approachable) Strombolian eruptive style [Chouet *et al.*, 1974; Blackburn *et al.*, 1976; Ripepe *et al.*, 1993]. The main focus of these observations has been on obtaining the best estimate of pyroclast exit velocity, an important eruption parameter linked to pressure in the conduit in particular and to eruption energetics in general. Moreover, the total volume, size distribution, and ejection dynamics (i.e., the evolution over time of ejection angle and velocity) have been addressed. More recently, also thermal camera observations [Patrick, 2007] and Doppler radar measurements [Seyfried and Hort, 1999; Hort *et al.*, 2003] have been used to investigate the ejection and settling of pyroclasts. Harris and Ripepe [2007] listed literature maximum ejection velocities at Stromboli volcano in the 22–72 m s⁻¹ range.

In the last decade, the higher acquisition rate and improved resolution of all the above mentioned techniques propelled relevant enhancements in the ability to track the motion of pyroclasts. The Doppler radar has been applied to the study of eruptions at Stromboli, Etna (Italy), Erebus (Antarctica), Colima (Mexico), Arenal (Costa Rica), and Santiaguito (Guatemala) volcanoes [Scharff *et al.*, 2008; Valade and Donnadieu, 2011; Gouhier and Donnadieu, 2011; Scharff *et al.*, 2012; Gerst *et al.*, 2013; Scharff *et al.*, 2015]. Beside a quantification of the velocity and size of pyroclasts, the Doppler radar analyses have revealed and measured the discontinuous nature of most such eruptions, discriminated the ballistic versus the ash plume component of the ejecta, and contributed to modeling the eruption dynamics and the associated geophysical signals (e.g., seismic and acoustic).

Commercial camcorders revealed eruption ejection velocity larger than 110 m s⁻¹ for small hydrothermal eruptions [Edwards *et al.*, 2017]. The high-speed thermal infrared imaging, with acquisitions at frame rates up to 200 frames per second, revealed ejection velocities at Stromboli volcano of up to 213 m s⁻¹, also illuminating the dynamics of the initial phases of the explosions and providing detailed velocity and grain size distribution of the pyroclasts [Harris *et al.*, 2012; Bombrun *et al.*, 2015]. Interestingly, thermal imaging of the trajectory of bomb-sized pyroclasts at Stromboli has revealed deviations from the predicted ballistic path, deviations that are interpreted as the result of pyroclast-pyroclast in-flight collisions [Vanderkluisen *et al.*, 2012]. These collisions, involving up to 10–15% of the analyzed trajectories, result in pyroclasts landing up to 4 times farther (or closer) from the vent with respect to the predicted position. The same authors also used oscillations in the cooling path of the observed pyroclasts to infer spinning, twisting, and tearing of the pyroclasts, suggesting that these processes may have a strong impact on the final landing point of the pyroclasts and the associated damage.

The high-speed visual imaging at frame rates of 500 or more frames per second has been also applied to explosive eruptions in this last decade. With respect to thermal imaging, visual imaging suffers from larger radiation attenuation by gas and vapor clouds but offers a wider variety of lenses and higher sensor resolution at a given frame rate. This last combination allows visualizing the motion of smaller and faster pyroclasts, with the result that exit velocities in excess of 400 m s⁻¹ have been measured in several Strombolian explosions at Stromboli volcano [Capponi *et al.*, 2016]. Beside the high velocity and accuracy of these measurements (error of less than 10%), the high-speed measurements highlighted time-dependent trends in the ejection velocity, angle, and size of pyroclasts that reflect eruption processes and allow estimating the associated masses and energies [Taddeucci *et al.*, 2012a, 2012b; Gaudin *et al.*, 2014]. The high-speed imaging of the first tens of meters above the vent of Strombolian explosions at Stromboli revealed that pyroclasts deceleration is strongly controlled not only by drag with the external atmosphere but also by the interaction of the pyroclasts with the gas jet that accompanies them [Taddeucci *et al.*, 2015a]. Extreme decelerations of up to 10⁴ m s⁻² occur at the front of the jet, where pyroclasts outrun the gas jet, while, below this front, pyroclasts travel in a reduced drag zone with slowly decreasing or even constant velocity. These direct observations also suggested that the reduced drag zone may vary over short timescales in unsteady eruption conditions.

3. High-Speed Observations

In the following, we illustrate original research characterizing the in-flight behavior of volcanic ballistic projectiles from different eruptions from several volcanoes worldwide.

3.1. Materials and Methods

High-speed videos used for this work were collected at the following volcanoes: Stromboli and Etna (Italy), Sakurajima (Japan), Fuego (Guatemala), Batu Tara (Indonesia), and Yasur (Vanuatu) (see Table 4 for details

Table 4. Main Features of the Analyzed Videos

Volcano	Acquisition Date (dd/mm/yy)	Video Name	Camera Type ^a	Resolution, $H \times V^b$ (Pixel)	FOV, $H \times V^b$ (m)	Pixel Size (m)	Camera-Vent Distance (m)	Frame Rate (fps) ^c	VBP ^d
Yasur	10/07/11	ya11	OPT	1024 × 1280	12.8 × 16.0	0.013	268	500	1
Yasur	10/07/11	ya15	OPT	1024 × 1280	12.8 × 16.0	0.013	268	500	2
Yasur	12/07/11	ya45	OPT	1280 × 1024	43.3 × 34.6	0.034	326	500	2
Stromboli	25/05/15	st27	NAC	1920 × 2560	43.9 × 58.5	0.023	374	250	14
Stromboli	27/05/15	st21	OPT	1024 × 1280	17.7 × 22.1	0.017	370	500	5
Stromboli	27/05/15	st54	OPT	1280 × 1024	22.1 × 17.7	0.017	370	500	4
Stromboli	26/05/16	st57	NAC	1920 × 2560	14.5 × 19.4	0.008	275	500	60
Etna	16/07/14	et12	OPT	1024 × 1280	25.5 × 31.9	0.025	240	500	20
Etna	16/07/14	et20	FLR	240 × 640	87.9 × 234.3	0.366	280	100	4
Fuego	13/01/12	fu09	OPT	1024 × 1280	39.5 × 49.3	0.039	826	500	2
Fuego	13/01/12	fu16	OPT	1024 × 1280	39.5 × 49.3	0.039	826	500	1
Fuego	13/01/12	fu23	OPT	1024 × 1280	39.5 × 49.3	0.039	826	500	1
Fuego	13/01/12	fu33	OPT	1024 × 1280	46.3 × 57.9	0.045	969	500	1
Fuego	13/01/12	fu36	OPT	1024 × 1280	46.3 × 57.9	0.045	969	500	1
Fuego	14/01/12	fu29	OPT	1024 × 1280	102.8 × 128.5	0.100	968	500	1
Fuego	14/01/12	fu39	OPT	1280 × 1024	128.5 × 102.8	0.100	968	500	1
Fuego	14/01/12	fu031	FLR	640 × 240	428.1 × 160.6	0.669	968	100	8
Fuego	14/01/12	fu039	FLR	640 × 240	428.1 × 160.6	0.669	968	100	10
Batu Tara	04/09/14	ba46	OPT	1280 × 1024	68.7 × 55.0	0.057	1217	500	4
Batu Tara	05/09/14	ba36	FLR	480 × 640	227.2 × 303.0	0.504	1217	50	10
Batu Tara	05/09/14	ba001	FLR	480 × 640	227.2 × 303.0	0.504	1217	50	10
Batu Tara	06/09/14	ba003	FLR	480 × 640	227.2 × 303.0	0.504	1217	50	5
Sakurajima	15/07/13	sa21	FLR	480 × 640	725.7 × 967.6	1.512	3673	50	6
Sakurajima	16/07/13	sa49	OPT	1024 × 1280	175.5 × 219.4	0.171	3673	500	7
Sakurajima	16/07/13	sa020	FLR	480 × 640	725.7 × 967.6	1.512	3673	50	5
Sakurajima	16/07/13	sa030	FLR	480 × 640	725.7 × 967.6	1.512	3673	50	5
Sakurajima	17/07/13	sa007	FLR	480 × 640	725.7 × 967.6	1.512	3673	50	8
Sakurajima	19/07/13	sa001	FLR	480 × 640	725.7 × 967.6	1.512	3673	50	7
Sakurajima	19/07/13	sa003	FLR	480 × 640	725.7 × 967.6	1.512	3673	50	8

^aOPT: Optronis CR 600x2; NAC: NAC Memrecam HX6; FLR: FLIR 655SC thermal infrared camera.

^b $H \times V$: Horizontal and vertical.

^cFrames per second.

^dNumber of tracked projectiles.

of video acquisition). Together, these volcanoes typically cover eruption styles and magma compositions ranging from Strombolian to Vulcanian and from basaltic to andesitic, respectively, with maximum height reached by projectiles spanning from a few tens to several hundreds of meters. Since our focus is on the behavior of individual projectiles and not on eruption dynamics, we redirect the reader to the literature for volcano-specific eruptive features. We filmed in-flight VBPs using a variety of cameras and settings, allowing spatial resolutions as high as 8 mm per pixel and field of view (FOV) as large as about 1 km across (Table 4). Close-ups have been used to detail projectile size and shape and the respective changes over time, while broader field of view allowed tracking VBPs motion over longer distances, often along their whole trajectory. In a limited number of cases we have been able to record the same event with multiple cameras. In these cases, we identified the same VBP in the two videos, using wider and narrower views to constrain its trajectory and size, respectively.

More than 200 individual projectiles have been tracked in the digital videos either manually or automatically (by using the MTrackJ and Tracker software packages [Meijering *et al.*, 2012; <http://physlets.org/tracker/>]) (Figure 6). Tracking was completed at a subpixel precision, so the error in position is less than the pixel size in each video. However, both automatic and manual tracking are subject to occasional, abrupt shifts in the tracking position when, e.g., the tracked pyroclast is temporary covered (by clouds or other pyroclasts) or it divides in smaller fragments. Tracking has allowed reconstructing the trajectory of each projectile within the camera FOV (see supporting information Table S1 for details of all tracked VBPs).

In all our recordings the high-speed camera was stationary (no panning). The in-flight behavior of VBPs, including the size and shape and their changes over time, has been investigated and documented cutting from each video frame a square area centered on the projectile and assembling all the areas in a small

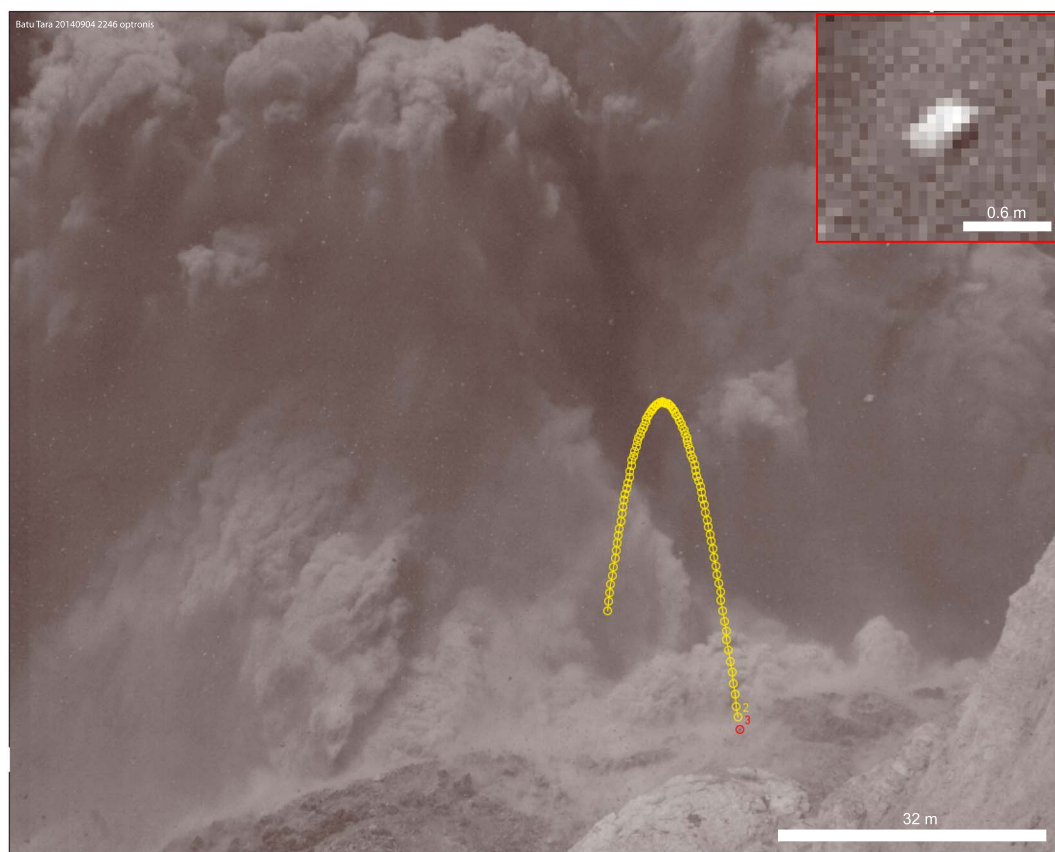


Figure 6. Still frame from a high-speed video of an explosive eruption at Batu Tara volcano (Indonesia), with superimposed the projected trajectory of a volcanic ballistic projectile (zoomed-in in the inset) reconstructed by manual tracking. The yellow circles represent the position of the VBP at time intervals of 40 ms (every 20 frames), for a total covered traveltime of 4.04 s.

video which, effectively, mimics the result of a camera tracking the projectile (Figure 7 and supporting information Movie S1). The result is a detailed reconstruction of the motion and in-flight dynamics of projectiles in the part of the flight path recorded by the high-speed video.

From these analyses we produced both the VBPs data set used to quantify in-flight processes and the illustrative movies provided as supporting information. The data set includes, for the tracked VBP, quantitative information on its trajectory, velocity, acceleration/deceleration, size, and occasionally other features such as rotation rate, collision, or ground-impact velocity and peculiarities when present. Velocity and acceleration were calculated from the projected VBP displacement over time, after smoothing with locally weighted least-square-error method [e.g., Cleveland, 1979] to minimize the propagation and amplification of the occasional shifts and small tracking errors into the calculation. For the size, we took advantage of the ubiquitous rotation to measure the minimum and maximum apparent axes of VBPs over their visible trajectory.

In the following, we describe and quantitatively parameterize several in-flight behaviors and properties of VBPs using the more than 200 detailed measurements present in the data set, together with many hours of observation of the tens of high-speed videos collected over the years.

3.2. In-Flight Size, Shape, Deformation, and Fragmentation of Volcanic Ballistic Projectiles

Concerning size, VBPs larger than 1 m have been observed at all volcanoes, weak Strombolian activity at Etna featured a maximum size of 3.3 m, and a record VBP of 7.7 m across was observed at Batu Tara. However, we note that the near-vent area at Sakurajima and Fuego volcanoes is both hidden from the camera and covered by ash during explosions, possibly hiding from view the largest projectiles. The measured VBPs span 0.05–0.73 m in equivalent diameter (diameter of a circle of area equal to the mean

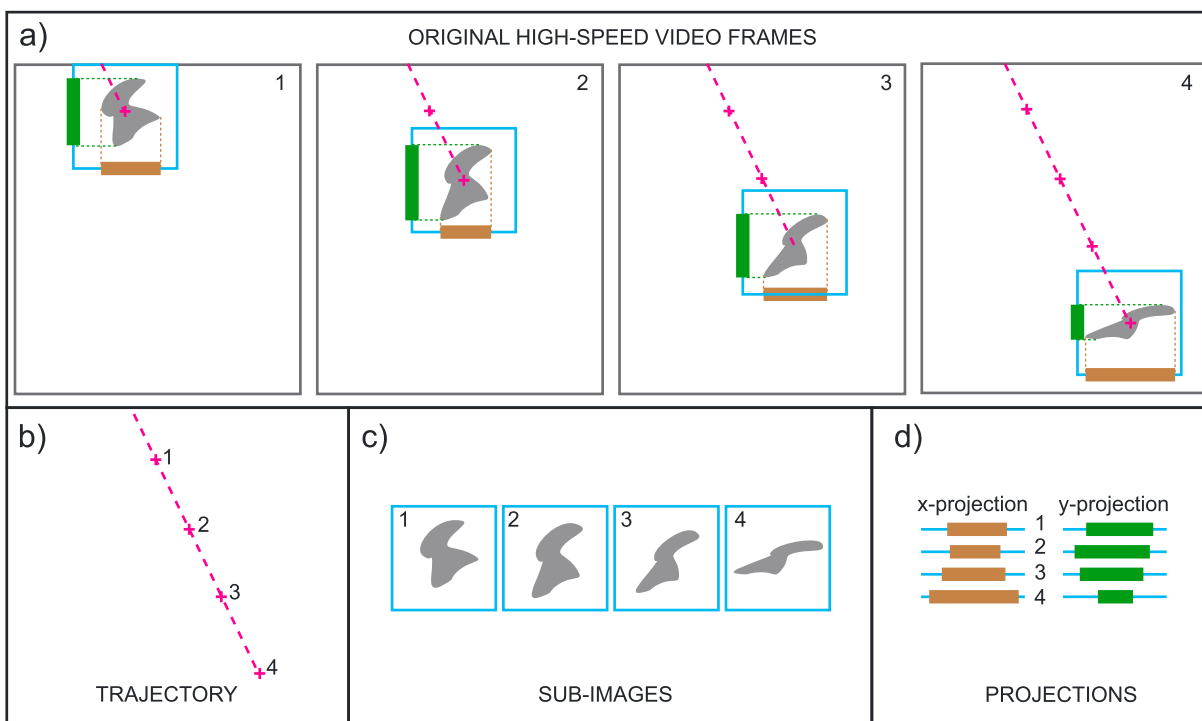


Figure 7. Example of high-speed video analysis and processing. (a) The center of a falling volcanic projectile is tracked over four (numbers one to four) subsequent frames of a high-speed video. (b) The track coordinates at the four different times define the projectile trajectory. (c) A montage of the small areas centered on the projectile provides a video centered on the projectile (see supporting information movies) and allows measuring its area (in grey). (d) Projection of the projectile over horizontal and vertical lines provides apparent projectile size at any time, used to measure projectile deformation and/or rotation over time.

projected area of the VBP), and 0.06–2.10 m and 0.03–0.54 m in long and short axis, respectively. The VBPs aspect ratio (length of the longest axis divided by length of the shortest one) ranges 1.27–14.10, and, in most cases, seems to increase with increasing projectile size, demonstrating that larger projectiles display, in general, more irregular shapes. The VBPs speed, spanning 2–229 m s^{−1} in absolute value (i.e., irrespectively of rising or falling direction), has an upper boundary which is roughly inversely proportional to the power of VBP size (Figure 8).

The in-flight shape of VBPs varies as a function of their source eruption and size. The VBPs in Strombolian explosions are for the most part ductile bombs, rigid blocks being subordinate. The projectile shape ranges from roundish-equant to highly irregular, including (i) very elongated, sinuous shapes with or without a larger bulge or protuberance at one extremity, (ii) “bolas” bombs (after the traditional Argentinean throwing weapon), with two or more bulges or irregular protuberances connected by thin, cylindrical “ropes,” (iii) amoeboid morphologies with multiple protuberances, and (iv) variably regular circles (Figure 9 and supporting information Movies S2–S8). Spheroidal to slightly elongated or bilobate shapes dominate centimeter-sized bombs, while larger ones tend to have more irregular shapes.

Bombs larger than a few centimeters are almost invariably observed to change shape during flight. The deformation is variable also in function of bomb shape, being more evident for more irregular shapes and less for the rounded ones. The elongated, sinuous shapes often twist in flight, connected bulges may swing around a variable position, and the protuberances of amoeboid shapes may swirl and flap (supporting information Movies S9–S14). There is no clear link between flight direction and in-flight orientation or deformation of the bombs, at least in the limited field of view of the high-speed cameras. In elongated bombs, the presence of a bulge occasionally controls deformation, and while the bomb is flying with the bulge at its front, the trailing, narrow body gets stretched. However, in bombs with the same morphology, the narrow body sometimes appear to swing and twist from behind to in front of the bulge. The clear formation of spindle-shaped bombs (or simply spindle bombs) has not been observed. In the closest cases, we observed rather elongated, irregular shaped bombs flying with their long axis variably oriented with

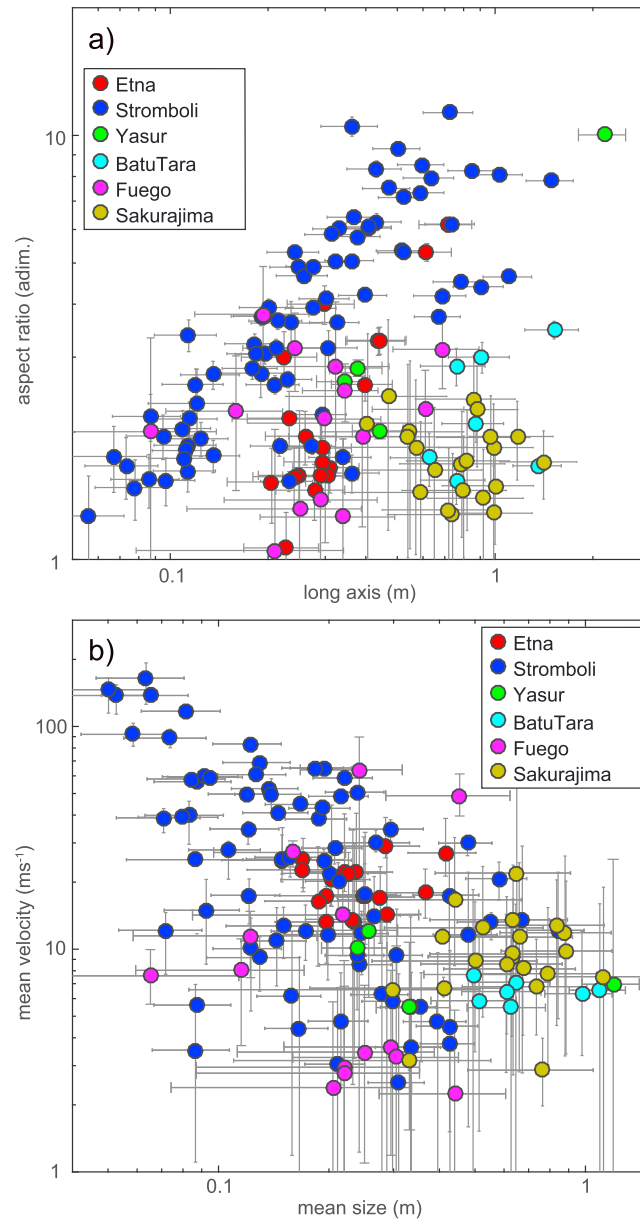


Figure 8. An overview of the key physical parameters of the filmed VBPs. (a) Relationship between the length of the long axis and the aspect ratio (ratio between the length of the long and short axis) of the analyzed VBPs, divided by source volcano (note logarithmic axes scale). (b) Relationship between the mean velocity (absolute value, averaged over the whole trajectory) and the mean size (average of long and short axes) of the analyzed VBPs. Error on size measurement (both mean size and long axis) includes ± 1 pixel to account for manual measurement error and an additional $\pm 15\%$ of the value to account for error in pixel size (function of the camera-VBP distance). Error bar in velocity covers the entire range from the minimum to the maximum velocity (absolute value) reached by each VBP. Data points with large velocity error bars mark VBPs tracked along both ascending and descending parts of their trajectories.

respect to the flight direction but then getting perpendicular to it. At this point, one of the two extremities got stretched and became narrower than the central part of the bomb (supporting information Movie S15).

Often VBPs are observed to fragment in-flight with no intervening collision. Most commonly, fragmentation is the extreme consequence of ductile deformation and can be classified in three main categories. First, stretching and tearing is observed to occur at the trailing edges of bombs, especially when this is the narrower part of them. In such cases, one or more small fragments detach from a main, elongated body. Second, twisting is observed to involve the body of elongated bombs and the appendices of amoeboid-shaped bombs in flight. Twisting appears to occur without much thinning of the bomb near the detaching point, and the resulting fragments are often a considerable fraction, up to a half, of the parent bomb. Third, bulges connected by thin “wires” or ropes are observed to increase their distance until the connecting portion became so thin that it fails, with timescale of the process shorter than 0.06 s. The remaining stretched appendices attached to the bulges are very thin and mobile, and their flapping motion occasionally may break them or bring them to merge with the rest of the bomb (Figure 9 and supporting information Movies S9–S14).

With respect to their Strombolian counterpart, our observations of Vulcanian-style eruptions are more limited in resolution (due to the longer filming distance). However, it is still clear that most of the observed VBPs are blocks with equant or slightly elongated shapes and aspect ratios (long to short axis ratio) usually smaller than two (supporting information Movies S16 and S17). Relatively large (>1 m) blocks appear, at least within our resolution range, to display rather rounded corners. Commonly, we did not observe blocks deforming during their flight. Only in one case we observed the

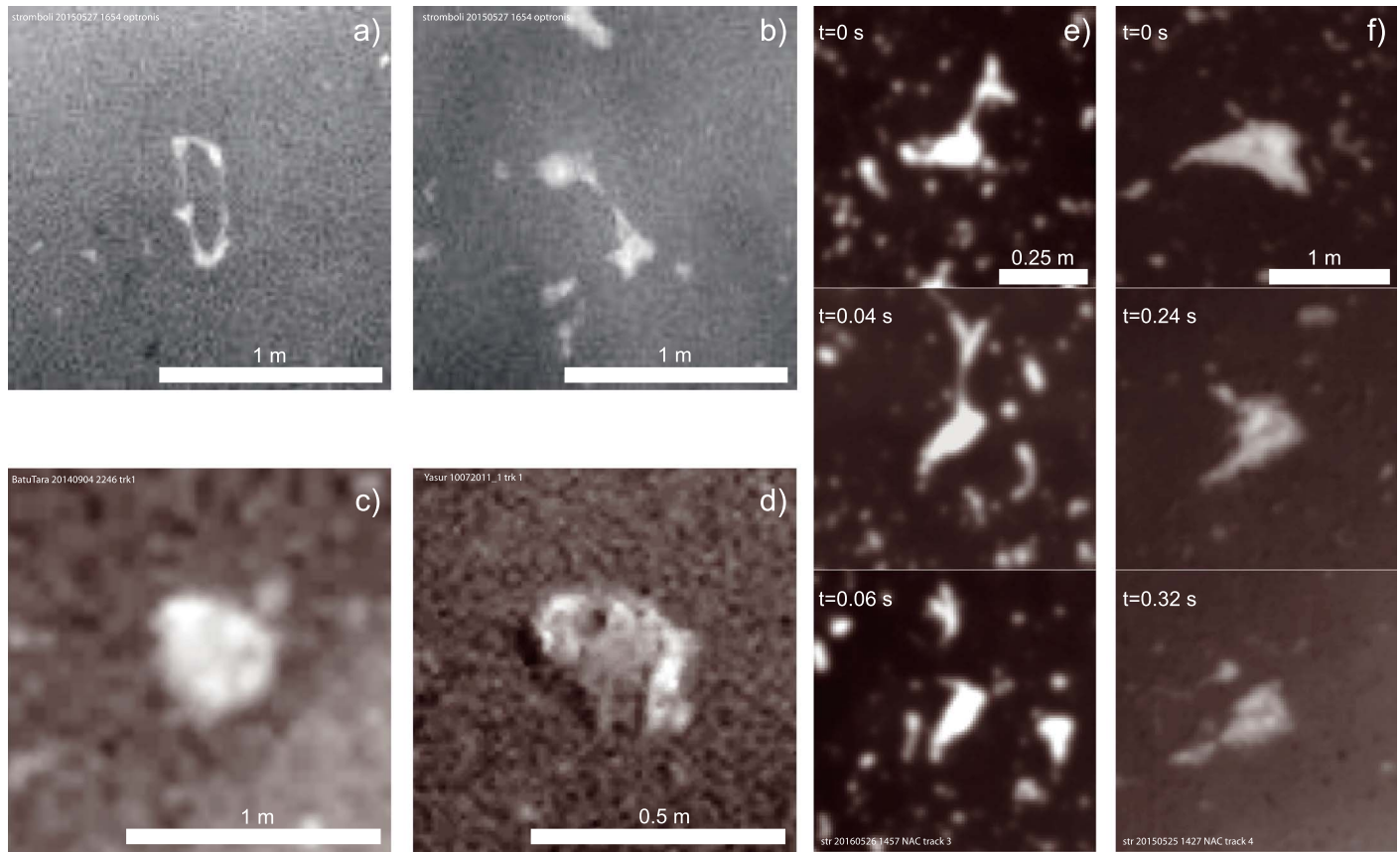


Figure 9. In-flight images of VBPs. (a and b) Examples of ductile bombs: a “hula-hoop” bomb and a bolas bomb, featuring two bulges tied by a narrow segment (supporting information Movies S3 and S4). Such irregular-shaped pyroclasts are unlikely to be preserved in the geological record. (c and d) Examples of brittle blocks with relatively equant shapes and rounded edges. (e) Sequence of three still frames illustrating the stretching and tearing of the thin neck between the two parts of a rotating bomb (time is relative to the first frame). Note the short duration of the tearing process (supporting information Movie S12). (f) Also in this case the VBP is rotating, but the two extremities detach almost simultaneously without any visible deformation. In the rest of the video the sharp truncations of the VBP extremities are still visible (supporting information Movie S18).

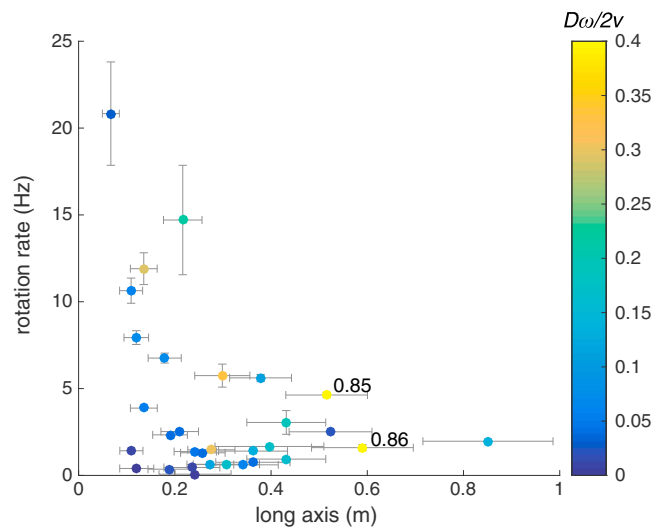


Figure 10. In-flight rotation rate of representative VBPs as a function of their size, expressed by the maximum length of their long axis. In color scale, the ratio $D\omega/2v$ (dimensionless), used to calculate the magnitude of the Magnus effect (numbered points are out-of-scale from the color scale). Error in the rotation rate is $\pm$ the time interval corresponding to four frames per 90° rotation.

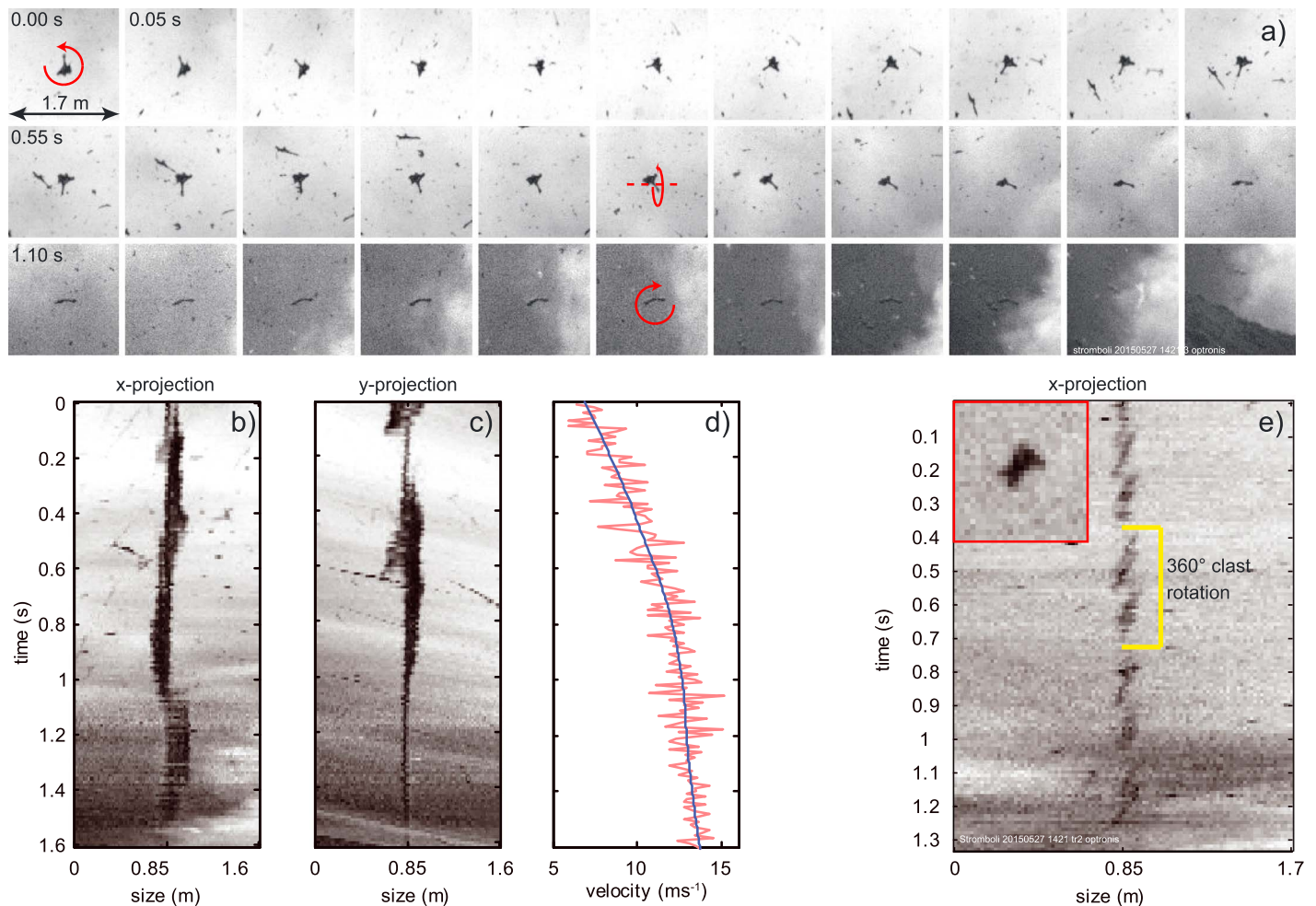


Figure 11. (a) Still frames (at 0.05 s intervals) illustrating the complex rotation of an irregularly-shaped VBP around at least two different axis. The peculiar shape of the projectile is visible thanks to the multiple rotations and, in turn, allows recognizing them. (b and c) Time evolution of the projection of the same VBP along a horizontal and a vertical line, outlining the rotation rate and changes in frontal area (see Figure 7 for an explanation of the projections). (d) Projected settling velocity of the VBP as a function of time (in red the original data and in blue the smoothed ones). Note how, starting from time 1.0 s, the projectile starts to attain its largest frontal area and its acceleration sharply decreases. (e) Horizontal projection of a VBP (shown in the inset) while spinning helicoidally at about 2.7 rps.

fragmentation of a pyroclast without any visible prior deformation: a crescent-moon shaped pyroclast lost both extremities in less than 0.1 s, without any deformation of the clast itself or of its extremities (Figure 9 and supporting information Movie S18).

3.3. Spinning and Rotation

Spinning and rotating are additional in-flight behaviors of VBPs which we observed ubiquitously. Measured rotation rates for a small subset of the best detailed VBPs (25 projectiles from two explosions at Stromboli and one each from one explosion at Yasur and Batu Tara) reveal a maximum rate of 20.8 rps (or Hz) and a mean value of 0.8 Hz. Despite the relatively limited number of samples, our data point to an inverse, nonlinear relationship between the maximum rotation rate and the size of VBPs (Figure 10).

Most observed pyroclasts display very unsteady rotation patterns. In fact, the videos show VBPs that rotate along multiple axis that shift over time, with changing rotation velocity and occasional, or even multiple, inversions in the direction of rotation (supporting information Movies S16, S17, and S19–S23). Occasional fast spinning is observed along one single rotation axis or even in a helical mode (Figure 11 and supporting information Movie S24).

Joint observation and measurement of VBP rotation and velocity illustrate well how, in VBPs with irregular shapes, rotation strongly influences velocity. For example, an irregularly shaped, rather platy VBP is

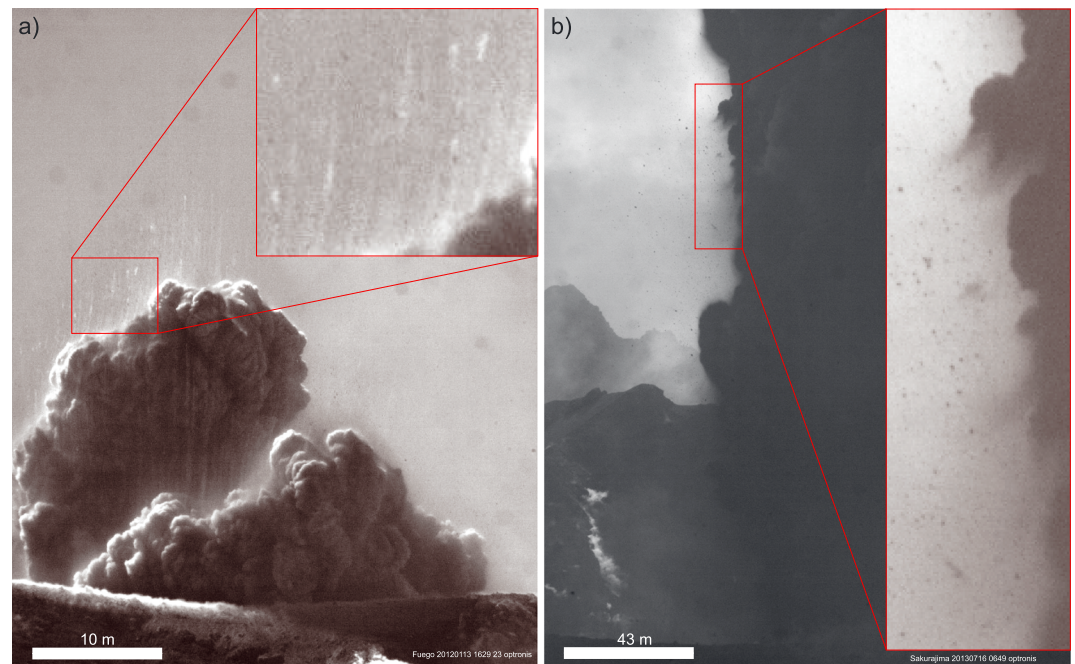


Figure 12. Still frames from high-speed videos showing in-flight collisions between VBPs. Collisions release the fine ash puffs visible in the images. (a) An explosion at Fuego volcano. Rising VBPs from subsequent ejection pulses collide with one another. The still picture captures the moment of maximum collision rate, when VBPs from a later pulse overtook those from a previous one. (b) An explosion at Sakurajima volcano. In this case, VBPs rising vertically from a new ejection pulse collide with ones that, moving downward and leftward, are leaving the eruption plume visible in the right-hand side of the image.

observed to reduce its downward acceleration from 6.1 to 0.8 m s^{-2} as its frontal area (area orthogonal to the travel direction) increases due to rotation (Figure 11 and supporting information Movie S25).

3.4. In-Flight Collisions

In-flight collisions between VBPs traveling at different velocity and direction are almost ubiquitous. Despite their common occurrence, which has been observed in all analyzed videos, it is remarkable that collisions do not occur at a constant rate during one explosive event. Rather, their timing and location is associated with the features and occurrence of ejection pulses at the vent [see, e.g., *Taddeucci et al.*, 2015b]. The most commonly observed case is that of collisions between rising VBPs from subsequent pulses. In this case, the fast projectiles ejected at the beginning of a later pulse overtake and collide with the slower ones ejected in the later phase of a previous pulse (supporting information Movie S26). This type of collision is mostly observed at the beginning of an explosion. We also observed rising pyroclasts colliding with falling ones, mostly at later times during one explosion (supporting information Movie S27). In both cases, the high rates of collision are observed to occur in waves, as the front of the newly ejected VBPs reach the cloud of already flying ones (Figure 12). While we observed rising pyroclasts colliding with both slower rising ones or with falling ones, collisions between two falling pyroclasts were not observed.

In-flight collisions result in a variety of consequences on the VBPs, as a function of several parameters, first of all their rheological/mechanical properties. During Vulcanian eruptions, for instance, the most commonly observed outcome of collisions is elastic momentum transfer and changes in the trajectory (Figure 13 and supporting information Movies S26 and S27). In one example, two blocks of volume $1.7 \times 10^{-2} \text{ m}^3$ and $5.9 \times 10^{-3} \text{ m}^3$ collided while rising along the same projected trajectory. The collision increased the velocity of the smaller VBP from 22.5 to 33 m s^{-1} while the larger one experienced only a small drop in velocity (from approximately 27.4 to 27.0 m s^{-1} , the observed increase in its deceleration is likely due to a change in the flight direction toward or away from the camera). Collisions always result in the release of ash puffs, possibly either by freeing ash present on the projectile surface or by fragmentation of the same surface. Only in a few, uncertain cases VBPs were observed to break in a few large clasts or to get entirely pulverized.

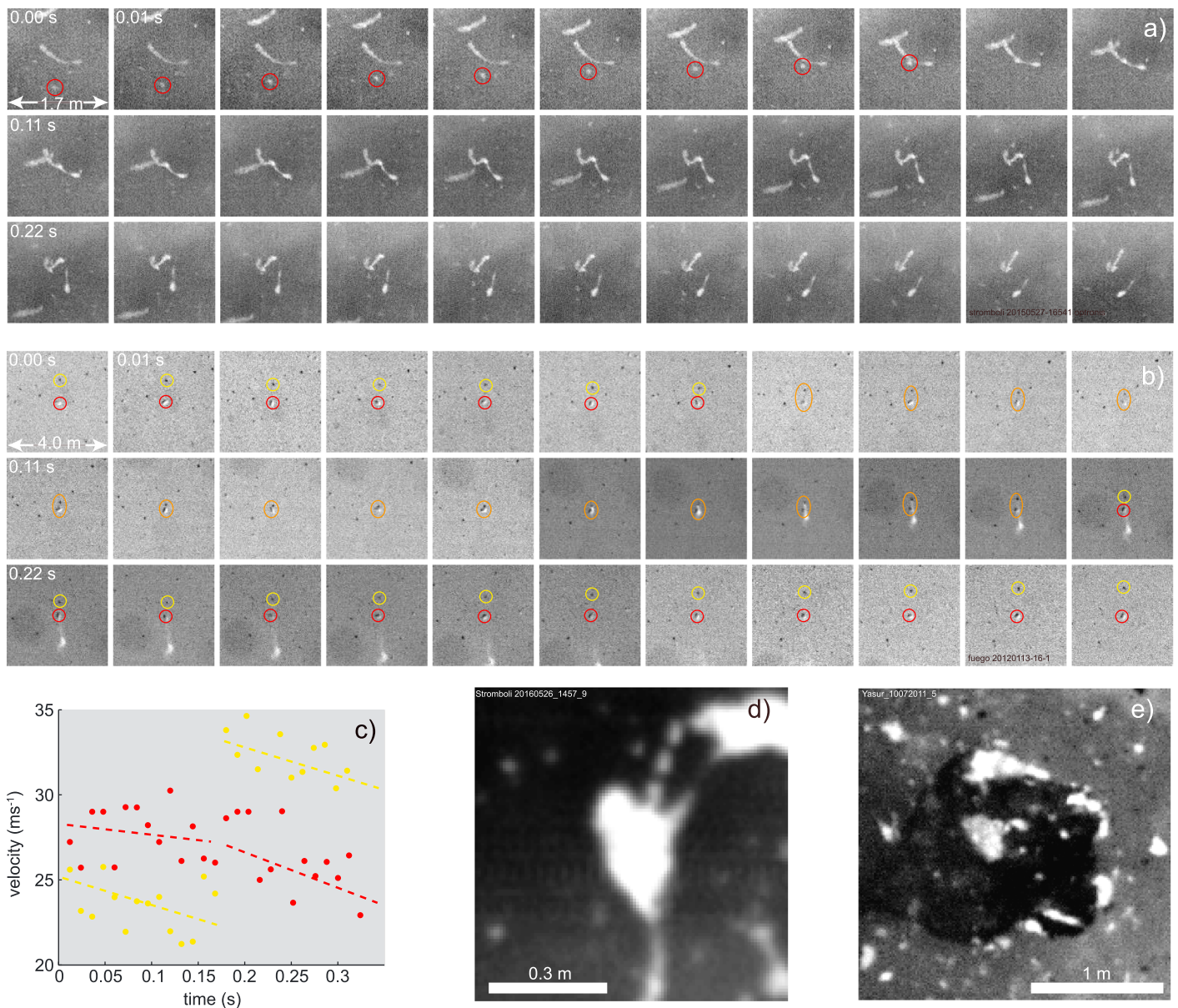


Figure 13. Still frames (at 0.01 s intervals) exemplifying two different outcomes of in-flight collisions between rising projectiles (the frames are centered on one of the two projectiles). (a) A plastic bomb deforming and fragmenting during a prolonged impact (supporting information Movie S28). (b) The rapid, elastic momentum transfer between two brittle blocks (supporting information Movie S26). (c) A plot of velocity over time for Figure 13b, the collision being marked by the abrupt change in the velocity of both projectiles at time 0.17 s (dashed lines: linear fits to velocity before and after collision). (d) A hot projectile being stretched and torn (note the line of droplets above the central pyroclasts) by the upward impact of a smaller one (supporting information Movie S31). (e) A flying, large block (dark mass) with multiple, hot (bright tones) pyroclasts plastered on it (supporting information Movie S34).

During Strombolian eruptions, the most frequently outcome of a collision is the deformation of the two VBPs. Deformation, in turn, may results in the agglutination of the two bombs and/or in their tearing apart in smaller fragments (Figure 13 and supporting information Movies S28–S33). We also observe the agglutination of molten VBPs on colder, not deforming ones, as well as piercing of larger bombs by smaller, faster, and colder ones (supporting information Movie S34). In the case of deforming and agglutinating VBPs, changes in the respective trajectories are more gradual and less obvious than in their elastic counterparts, also because fragmented bombs quickly reorient after the collision according to the new shape and flight direction (supporting information Movie S35). Both ductile bombs and brittle blocks markedly increase their rotation rate after collisions.

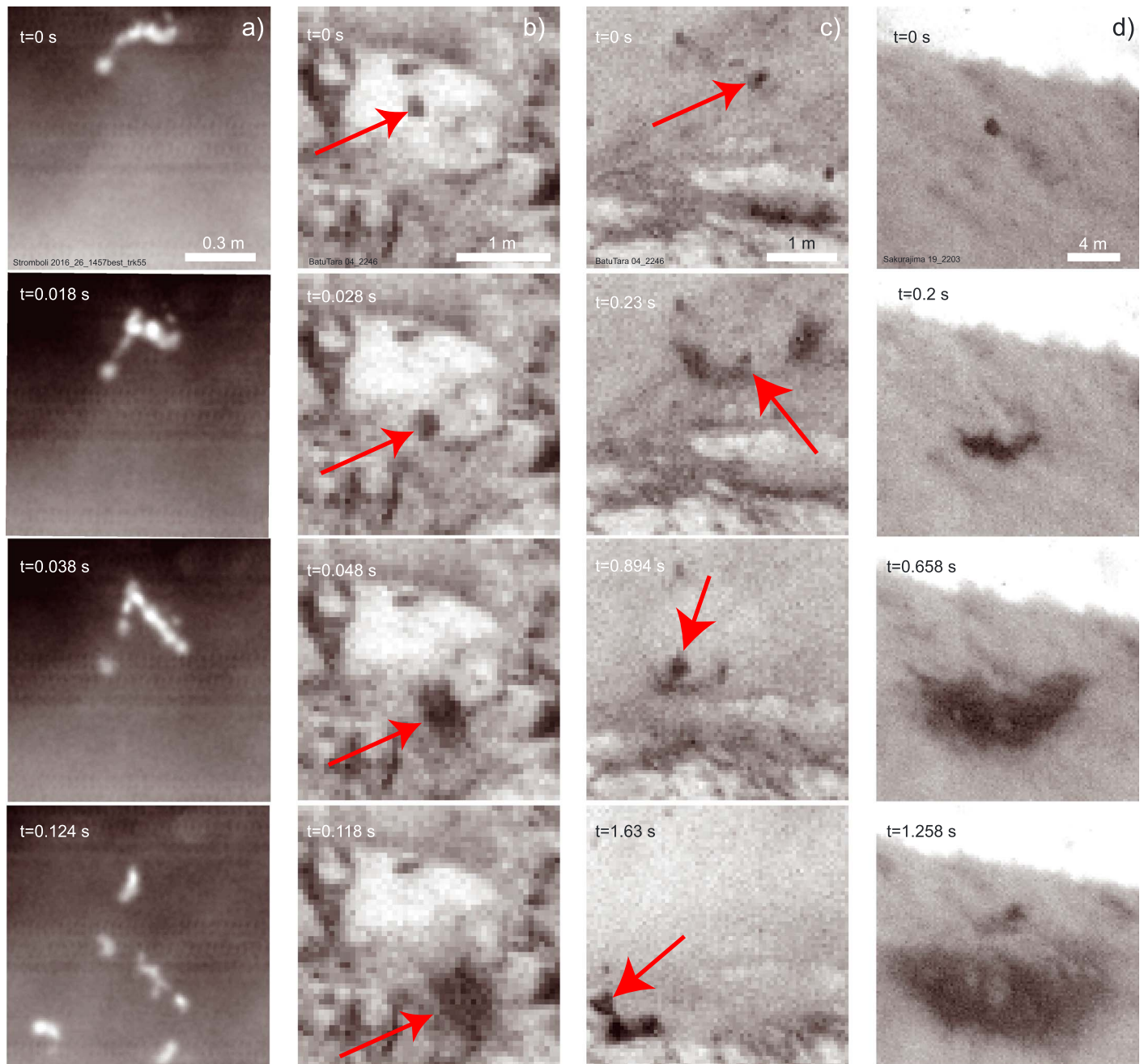


Figure 14. Ground impact of VBPs. (a) An elongated bomb being deformed and fragmented by the impact with the steep inner crater wall at Stromboli (supporting information Movie S37). (b) A block (red arrow) being finely fragmented by the impact with a hard, probably lava substratum at Batu Tara. (c) Multiple craters forming by the first impact and following rebounds of a single block (red arrow, still in flight at $t = 0$) impacting what can be assumed to be an ash/lapilli layer (darker gray area) covering a harder substratum (brighter color area outcropping in the lower part of the image, which is the same substratum as in Figure 14b) (supporting information Movie S40). (d) The development of an ejecta fan during craterization caused by the impact of a large block (still in flight at $t = 0$) with a soft substratum (likely ash/lapilli layers from previous explosions) at Sakurajima. Note the incipient collapse of the impact ejecta cloud at $t = 1.258$ s.

3.5. Ground Impact of Volcanic Ballistic Projectiles

Within our database we observed a number of processes related to the impact of VBPs with the ground. The nature of such processes varies, as in the case of in-flight collisions, with the mechanical/rheological state of the VBP and, in this case specifically, also with the nature of the impacted substratum, including its lithology (coherent rock versus loose granular material) and orientation relative to the VBP trajectory.

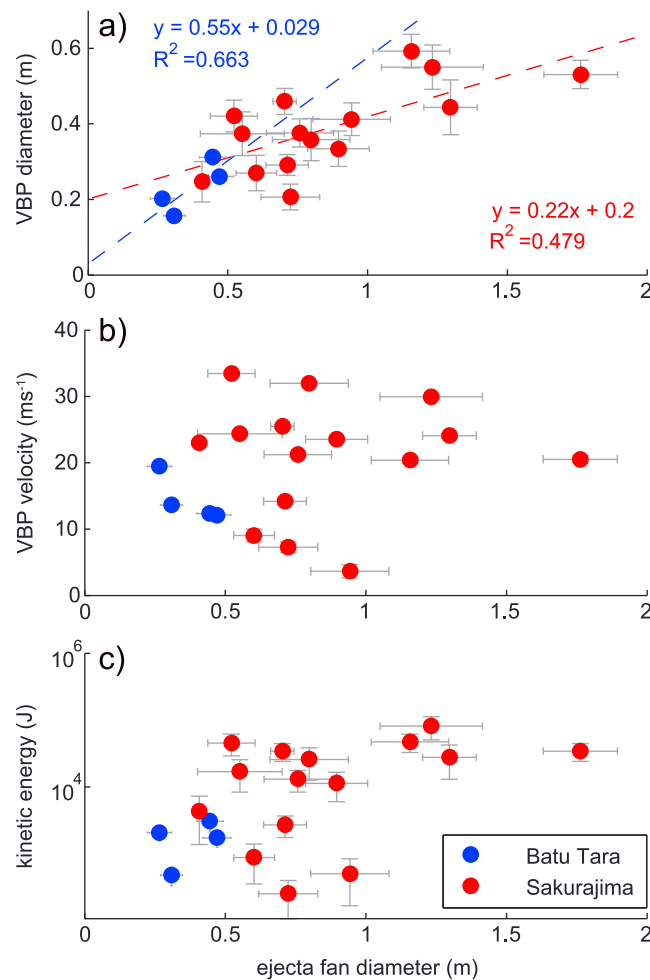


Figure 15. The size of the ejecta fan (proxy for crater size) as a function of (a) impactor size, (b) impact velocity, and (c) impact kinetic energy for VBPs at Batu Tara (blue dots) and Sakurajima (red dots) volcanoes. Dashed lines in Figure 15a are linear regression fits. Error bars are ± 1 standard deviation of repeated measurements (ordinate axis error bars in Figure 15b are smaller than symbols).

Of the observed ductile bombs, only a minority survive impact without breaking and none without deformation. In most cases impact results in tearing of the bomb, followed by relatively prolonged (seconds) deformation (supporting information Movies S36 and S37). This last process is more pronounced for impacts occurring on relatively steep slopes, where landed bombs often obtain rounded shapes and merge with one another while rolling downward (Figure 14 and supporting information Movies S38 and S39).

The impact of blocks often results in their fragmentation, especially for those with more complex shapes. Rounded blocks are observed to fragment only when impacting a hard substratum, i.e., a lava flow or boulder. In this case we observed both the breaking of the block in a few, smaller ones and its entire fragmentation, with the production of abundant ash (Figure 14).

Blocks and bombs impacting on a loose, granular substratum have been observed to produce an impact ejecta cloud, or fan, the unequivocal mark of the formation of an impact crater. The timescale associated with this process is in the order of 0.1 to 1 s, depending on projectile and substratum properties. Occasionally, the same block is observed to bounce out of the newly formed crater, bouncing and rolling downward, to form additional smaller craters (Figure 14 and supporting information Movie S40).

We have no direct information about the size of the craters formed by the observed VBPs impacts. However, from some of the videos we were able to measure the size of the ejecta fan and use this as a proxy for crater size for a limited number of cases (4 VBPs at Batu Tara and 13 at Sakurajima). The diameter of the ejecta fan has been measured at its base and in the first frames of its formation, as the best proxy for crater size (D_c). Scattering considered, the crater diameter grows approximated linearly with the VBP size, with different slope for the two volcanoes. There is no obvious relationship with impact velocity and a weak direct proportionality with impact kinetic energy (Figure 15).

3.6. In-Flight Trajectory of Volcanic Ballistic Projectiles

The trajectories of tracked VBPs, projected onto the view plane of the camera, cover a wide range of travel distances, durations, and intervening processes.

The longest trajectories we recorded follow the VBPs in their ascending and descending paths, from close to the vent to their landing point, with a lateral range and maximum elevation spanning about 5–250 and 10–500 m from the vent, respectively. Most of these trajectories display the typical, asymmetric curve of ballistic projectiles (Figures 16a–16c), occasionally modified by centimeter scale to decimeter scale

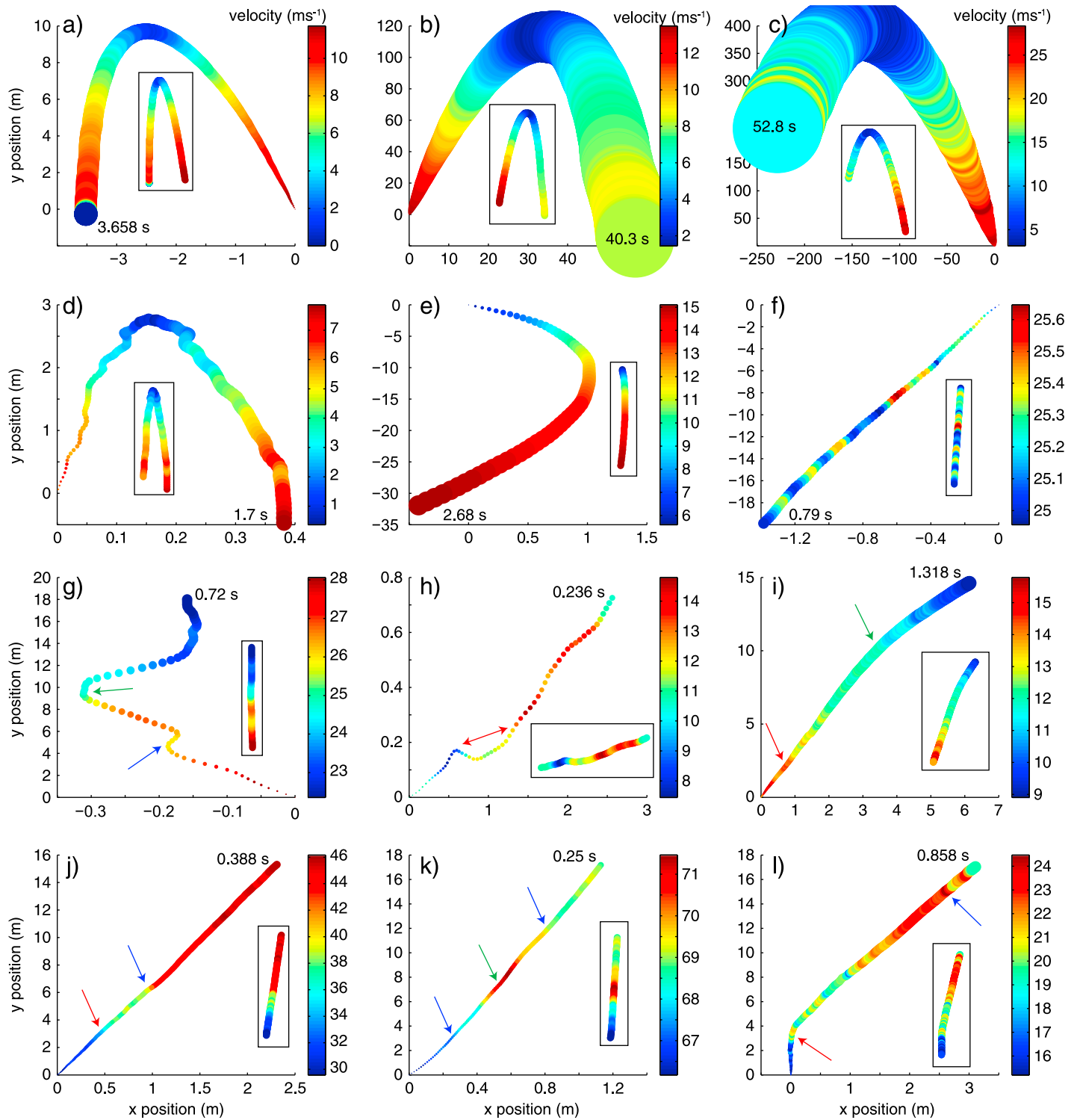


Figure 16. (a–l) Examples of projected VBP trajectories, expressed as projectile vertical (y) and horizontal (x) displacement (in meters) from the first tracked point. Note the differing x and y axis scales, used to highlight trajectory details (in insets, the undistorted trajectory). Color scale is proportional to projectile absolute velocity, while circle size increases with time (total track duration is reported next to the last tracked point). Red, blue, and green arrows mark the time of collision, tearing, and sudden deformation/reorientation of the VBP, respectively.

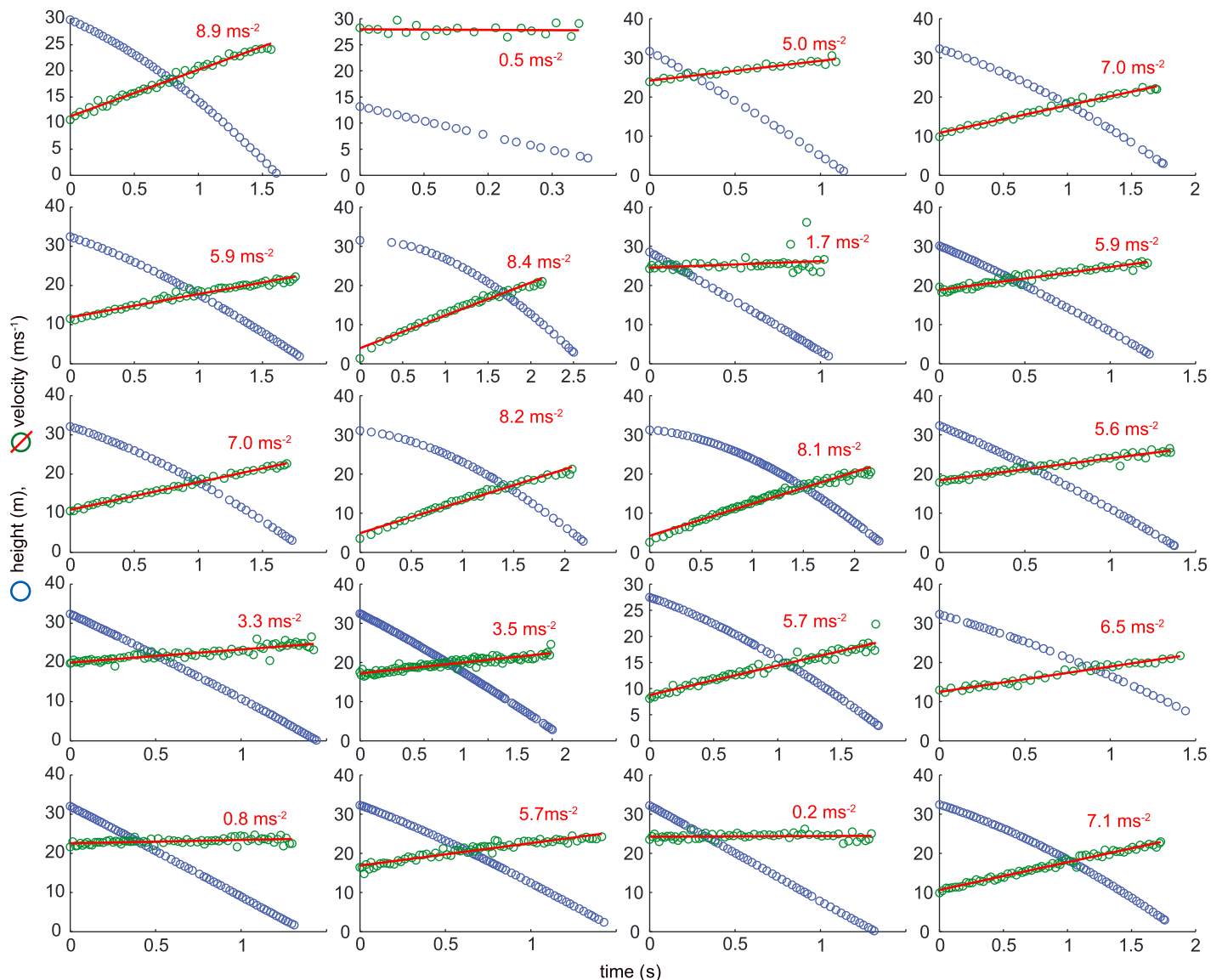


Figure 17. Time variation of height above the vent (blue circles) and velocity (green circles) (same scale) of selected, falling VBPs from explosions at Etna volcano. In red, the linear fit to the velocity data and the corresponding value of acceleration. Note that axes scale changes from plot to plot.

wobbling motions, particularly in the case of irregularly shaped projectiles (Figures 16d and 16g and supporting information Movies S3 and S6).

Conversely, most close-up videos captured only a small part of the ascending or descending trajectory of the VBPs. Descending trajectories are usually smooth in their path, with occasional lateral deviations caused by winds blowing transversally to the camera direction of view (Figure 16e). Descending trajectories mostly display a gradual increase in projectile velocity over time, and only the smallest VBPs are observed to fall at an almost constant velocity over distances of several meters. Even in such cases, however, velocity is often observed to display small-scale oscillations, in the order of 5%, around a mean, constant value (Figure 16f).

In contrast to descending trajectories, ascending ones are more irregular, both in their flight path and their velocity history. Smoothly decelerating trajectories are observed to be interrupted by processes such as (1) in-flight collisions, (2) projectile fragmentations, and (3) relatively abrupt changes in projectile shape due to deformation and/or reorientation (see also above, section 3.3). For instance, we observed one of the bolas bombs first jerking laterally by about 5 cm (clearly visible in the 8 mm/pixel resolution video) when a small part of it detached and then sharply changing its apparent flight direction by about 3° as it flipped and

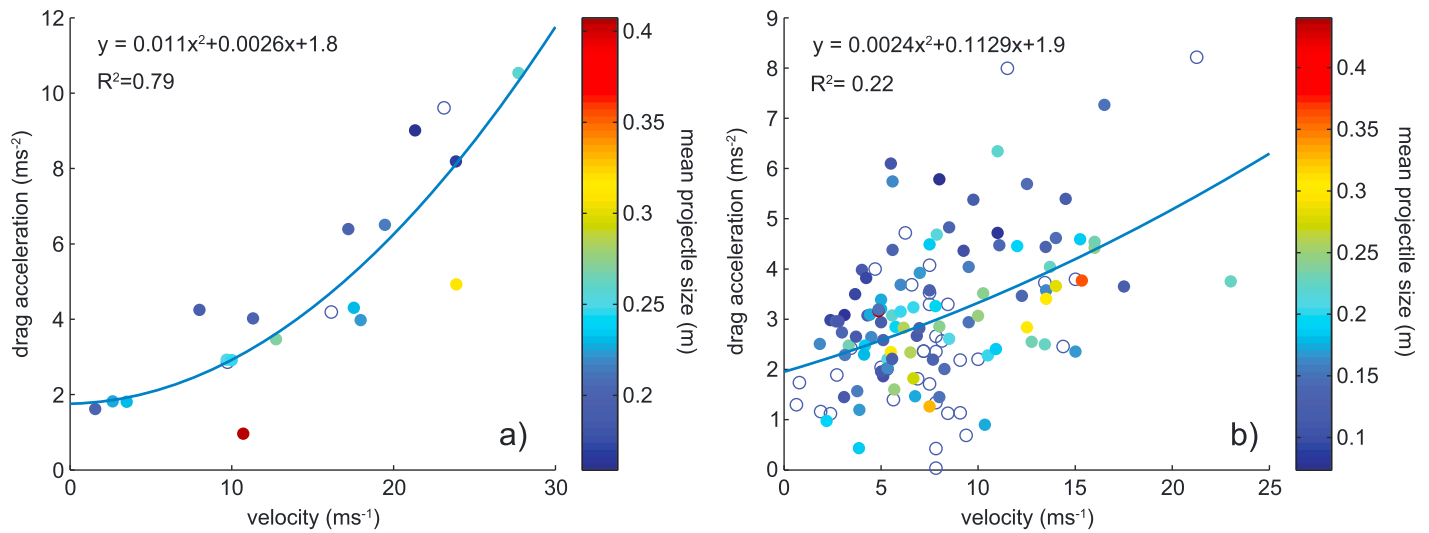


Figure 18. Changes in drag deceleration (i.e., gravity acceleration minus measured VBP deceleration) as a function of VBP fall velocity for selected falling VBPs from Strombolian explosions at (a) Etna and (b) Stromboli volcanoes. In color scale, the average size of VBPs (empty if not available).

rotated (Figure 16g and supporting information Movie S6). Collisions are observed to both accelerate and decelerate projectiles, according to the relative velocity differential and mass of the colliding VBPs (see above, section 3.4, Figures 16h and 16i, and supporting information Movie S41). Several projectiles are observed to accelerate, rather than decelerate, their ascending motion without any collision, as deduced by (i) the gradual increase in velocity and (ii) the lack in the video of the usual indicators of collisions, such as a collider, ash puffs, or changes in projectile shape. A bomb rising at about 32 m s^{-1} was observed to be first accelerated to $36\text{--}38 \text{ m s}^{-1}$ by a collision, and then to accelerate again to more than 40 m s^{-1} while being torn apart (Figure 16j), and similar acceleration and deceleration patterns are observed also as a consequence of VBP reorientation (Figure 16k). Interestingly, a bomb whose trajectory was deviated by a collision is observed to accelerate from 20 to 24 m s^{-1} without any other substantial change in its morphology or trajectory (Figure 16l).

3.7. Drag Force Acting on Flying Volcanic Ballistic Projectiles

Air drag was quantified for a limited set of pyroclasts, some of them rising and some other falling, from Strombolian explosions at Etna (20 projectiles), Stromboli (138 projectiles), and Yasur (5 projectiles), avoiding colliding or deforming VBPs. Over the short tracking timescale ($t < 2 \text{ s}$), most projectiles show an almost linear change of velocity over time, i.e., a constant acceleration, with values ranging from 8.9 m s^{-2} , close to gravity (9.8 m s^{-2}), to 0.2 m s^{-2} (Figure 17).

From the above plots we extracted the drag acceleration (D_g) of each VBP, i.e., the difference between gravity acceleration pulling downward and the actual, measured vertical acceleration, and the absolute magnitude of the velocity at the beginning of the measurement (V_o). As predicted by equation (16a) or (16b), in general, D_g increases with V_o and with decreasing projectile size (for any given velocity), but the scatter is rather large (Figure 18).

We used a number of selected VBP trajectories to attempt a field-based estimate of their drag coefficient. The selected trajectories belong to projectiles not affected by collisions and with minimal deformation, rotation, and other intervening dynamics and display the smallest errors in size and velocity. The apparent drag coefficient for VBPs either falling or rising can be estimated considering only gravity and drag force from either equation (16a), considering only horizontal wind component u_x , or (16b), considering only vertical wind component u_z , as follows:

$$C_d = \frac{4\rho_b D}{3\rho_a} \left(\frac{-a_x}{(v_x - u_x)|\mathbf{v} - \mathbf{u}|} \right) \quad (45a)$$

$$C_d = \frac{4\rho_b D}{3\rho_a} \left(\frac{-a_z - g}{(v_z - u_z)|\mathbf{v} - \mathbf{u}|} \right) \quad (45b)$$

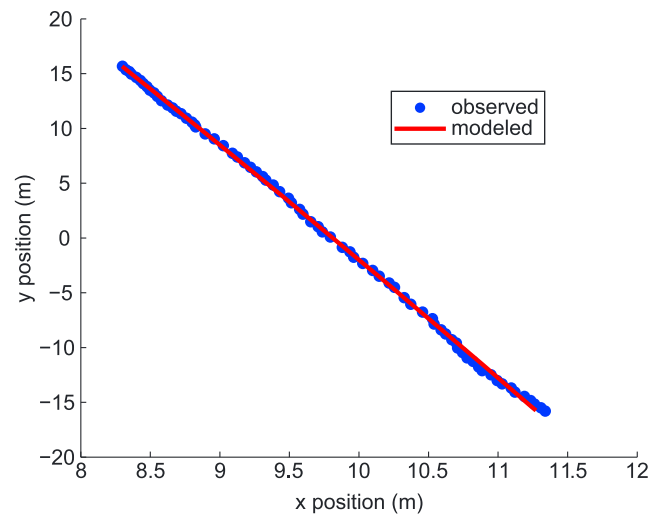


Figure 19. The trajectory of most of the projectiles selected for the drag force analysis can be fitted quite well considering (equations (16a) and (16b)) and the apparent drag coefficient values presented in Figure 20. This example corresponds to track 13 of Etna volcano with an apparent $C_d = 4.75$ and considering a horizontal wind in the field of view plane of 2.6 m s^{-1} .

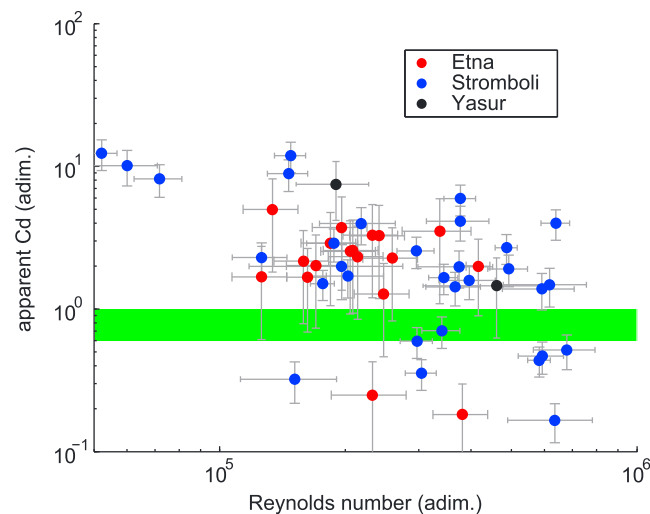


Figure 20. Apparent drag coefficient (C_d) as a function of Reynolds number (Re) for suitable trajectories of VBPs from Etna, Stromboli, and Yasur volcanoes. Each point corresponds to the average apparent C_d of an individual projectile. Re is calculated using (equations (22) and (23)) considering the corresponding volcano altitude. In the case of Etna projectiles, a sharp decrease of the C_d is observed at around $Re = 3 \times 10^5$ which might correspond to $Re_{critical}$. C_d values also decrease with Re for the Stromboli volcano, but scatter is larger. The errors were calculated for each particle using the variance formula for propagation of uncertainties [e.g., Bohm and Zech, 2014] for all the applied equations considering the uncertainties associated with their size (standard deviation of the size measurements of each particle in several frames), density (standard deviation for density values reported in the literature for Stromboli and Etna [Polacci et al., 2006; Polacci et al., 2009; Gurioli et al., 2014]), and velocity estimation (uncertainty associated with the slope fitting of the observed data). Note how C_d values from literature mostly range from 0.6 to 1.0 (green band).

For most of the short trajectories, it can be assumed that the acceleration remains constant throughout the video (Figure 17). Figure 19 shows that the trajectory calculated from equations (16a) and (16b) considering this assumption provides a remarkable fit to the observed data. In this case, the terms in brackets in equations (45a) and (45b) can be directly obtained from the trajectory data observed in the high-speed videos. It is important to note that $|\mathbf{v} - \mathbf{u}|$ actually refers to the relative velocity between the VBP and the ambient wind considering the three components (x , y , and z). This fact is particularly important considering that in general the plane observed with the camera does not necessarily coincide with the trajectory plane. Furthermore, since the data provide information solely on two directions, only two unknowns can be computed: drag coefficient and one component of the surrounding fluid velocity. For the most of the projectiles we considered a horizontal wind component (in the direction of the plane of view of the camera) and used equation (45a); however, we also observed in some rising particles near the vent that the upward velocity of the expanding gas controlled the drag force and calculate the drag coefficient accordingly (using equation (45b)). An important source of uncertainty comes from the lack of data on the ambient wind velocity vector (including its three coordinates), and partially for this reason we elected to term C_d values obtained with this procedure as “apparent” drag coefficient.

In order to calculate the apparent C_d values using equations (45a) and (45b) the mean VBP diameter was estimated as the geometrical average of minimum and maximum axis measured in several frames for each particle (its associated uncertainty is given by the standard deviation). Since the observed particles were not collected in the field, we considered a particle bulk density ($1100 \pm 246 \text{ kg m}^{-3}$, $1500 \pm 400 \text{ kg m}^{-3}$, and $2000 \pm 400 \text{ kg m}^{-3}$ for Etna,

Stromboli and Yasur, respectively) as reported in the literature [Polacci *et al.*, 2006; Polacci *et al.*, 2009; Gurioli *et al.*, 2014]. The apparent C_d calculated for the selected VBPs ranges 0.1–10 for Reynolds numbers ranging 5×10^5 to 10^6 , with no apparent difference between the different volcanoes (Figure 20).

4. Discussion

We caution the reader on the limited representativeness of our data set, which is circumscribed to small-scale eruptions that launch VBPs to relatively short ranges. Still, on these limited cases, it probably represents the most accurate analysis of VBP in-flight dynamics to date. On the whole, the observed dynamics suggest that there is no justification for the assumption, often used in modeling VBPs, that few forces other than drag act on them after they are ejected from volcanic vents. VBP trajectories are, in reality, more complex than those postulated by such assumptions, due to processes such as in-flight stretching, deformation and breaking, spinning, collisions, variable drag coefficient, and the influence of the jet dynamics near the vent. On these grounds, we draw first-order considerations on the observed dynamics and their implications for the interpretation of VBPs in volcanic deposits and in modeling their trajectory, eventually impacting VBP-related volcanic hazard assessment.

4.1. Stretching and Deforming

Grain size distribution is a key aspect of volcanic deposits, often related to magma fragmentation, and a proxy for eruption history [e.g., Cashman and Scheu, 2015]. Our observations suggest that current measurement of grain size distributions of pyroclasts from Strombolian activity may be biased toward more fine grain sizes, due to two reasons. First, the majority of flying bombs larger than a few centimeters display very irregular shapes, with larger masses separated by thinner bridges. Bombs with shapes such as “hula hoops” or bolas are very rarely observed on the ground and are unlikely to survive ground impact. The tapering ends of these bombs are also unlikely to be recognized as connections between larger bodies after ground impact, because still fluidal bombs may anneal sharp fracture surfaces, as we observe for landed bombs rolling downward. Second, in-flight tearing and fragmenting of bombs are also commonly observed in our videos. Also this process is likely to affect selectively larger bombs, due to their complex shape and larger surface exposed to drag. Both in-flight tearing and breaking on landing will eventually produce a grain size distribution of the deposit that is finer than that originally erupted at the vent, especially with respect to the coarse tail of the distribution (i.e., particles above several centimeters in size).

Complex shapes and in-flight deformation of VBPs imply unsteady drag forces, due to changes in the total surface and the frontal area of the projectile during its motion. They also imply that the center of mass of the projectile moves over time. The effect of these two factors and their combination on the flight trajectory of VBPs has not yet been explored and may be nonnegligible, potentially impacting hazard map accuracy. In general, a decrease in the size of VBPs and an increase in their surface area, both resulting from tearing and stretching, are expected to decrease their range.

Stretching and tearing has been invoked to explain observed anomalies in the measured surface temperature of bombs [Vanderkluyzen *et al.*, 2012], and our observations substantiate this explanation, with potential implications for the assessment of the thermal state of the pyroclasts. Stretching-induced changes in the thermal state of pyroclasts may also be important for the fragmentation of their surface into smaller, ash-sized fragments, in analogy with processes observed and modeled during phreatomagmatic eruptions [Mastin, 2007].

The thermal state of pyroclasts is also important to assess their rheological-mechanical properties. The observed in-flight deformation of VBPs, both ductile and brittle, opens the ways for potential in situ measurements of these properties at eruptive conditions. In the future this could be done by relating the computed stresses imparted by inertia and drag on the projectile with the observed deformation rates and fracturing, possibly obtaining information on the in-flight viscosity and tensile strength of pyroclasts.

4.2. Spinning

We observe rotation and spinning to frequently affect the trajectory of VBPs in several ways. Rotations change the frontal area of VBPs during flight, in turn changing the drag force exerted by air. Since most VBPs have irregular shapes, this effect is relevant in general and maximum for very oblate or platy morphologies (Figure 11). In such cases, the use of an average value of drag coefficient would be a viable

solution for trajectory modeling, if the projectile would expose cyclically the same frontal areas to the air flow. However, this is often not the case in our observations. Possibly due to the relatively short observed travel path, VBPs do not seem to attain any obvious equilibrium flight orientation, either static or dynamic, but rather to rotate along complex and counterintuitive paths.

Multiple causes can set projectiles in rotation, including, e.g., asymmetries in the initial acceleration, projectile shape, and collisions. Our observations are limited to outside the vent. There, VBPs are set in rotation in three visible cases. First, the in-flight collision with other projectiles, which in most cases initiates or enhances rapid spinning of both target and impactor projectiles. Second, the in-flight fragmentation of bombs by stretching, which evolves through the development of asymmetric shapes and rapid shifts in the center of mass of the system when detachment occurs. Third, rarely we observe VBPs starting, or enhancing, their rotation when entering an eruptive gas jet (see below). All these processes are enhanced by high gas and pyroclast ejection rates in the proximity of the vent, and we expect most VBPs to acquire most of their angular momentum in the first 10–20% of their flight path.

Previously reported rotation rates for VBPs at Stromboli averaged 2.18 Hz over 54 pyroclasts [Vanderkluyzen *et al.*, 2012]. Here on a smaller number of projectiles mostly from the same volcano, we found a mean and maximum values of rotation of 0.8 and 20.8 Hz, respectively. Despite the limited number of measurements, an inverse relationship seems to link maximum rotation rate and projectile size (Figure 10). Given that smaller projectiles require a smaller force to be set spinning, the observed relationship may reflect the maximum values of the forces providing angular momentum to the VBPs in the observed eruptions.

Spinning can curve the trajectory of VBPs according to the Magnus force (equation (8)). This force is always perpendicular to the relative velocity vector between the VBP and air and the rotation axis vector. As mentioned above, the rotation axis shifts over time in a very complicated manner that is very difficult to parameterize from the video observations. However, a maximum value of the Magnus force can be estimated considering the case when both vectors are perpendicular to each other. As a first-order approximation, the lift coefficient is estimated from equations (20) and (21) considering the data measured in the videos. If we further assume that the Magnus force remains constant during the short tracking timescale ($t < 2$ s), a deviation angle ranging between 1° and 23° with respect to the straight trajectory plane is theoretically estimated using equation (8) for the measured projectiles, and significant deviations (up to 9 m in a time window of 2 s interval) are theoretically expected for nearly vertical trajectories. Higher deviations during the full trajectory can be expected for all the projectiles even when the angle between the velocity vector and the rotation axis changes continuously. This simple analysis of the rotation observations clearly shows that the Magnus effect can be relevant to VBPs trajectories and can easily explain deviations in the order of 20–40° as observed by Waitt *et al.* [1995] at the ballistic field of the 1992 Mount Spurr eruption. This effect is particularly relevant for the determination of vent location studies from VBPs distribution. Further 3-D observations of VBP trajectories with two different cameras will be essential to quantify this phenomenon for volcanic projectiles.

4.3. Collisions

In all observed cases, the in-flight collision of VBPs is tightly linked, both for their timing and location, with the occurrence of ejection pulses at the eruptive vent, as already hypothesized by Vanderkluyzen *et al.* [2012]. During such pulses, a large number of pyroclasts is ejected in a very short time span and with a relatively high velocity, especially at pulse beginning [Taddeucci *et al.*, 2012a, 2012b]. Ejection pulses thus provide the high concentration of pyroclasts and the substantial velocity differential required to promote the collisions. Pyroclast ejection may vary in number, direction, and velocity during one pulse and between pulses [Gaudin *et al.*, 2014], and this variability affects collision variability. For instance, VBPs from two subsequent pulses may collide while rising, the fast pyroclasts ejected at the beginning of the later pulse overtaking the slower ones ejected toward the end of the previous pulse. This type of collisions is favored if projectiles are ejected vertically and with a narrow spreading angle. We also observed rising VBPs colliding with falling ones. In this case, shifts in vent position or ejection angle may favor the newly ejected pyroclasts to collide with those ejected previously, still granted a relatively narrow spreading angle at the vent. We never observed collisions between two falling VBPs, probably because (i) velocity differentials in the falling projectiles, induced by drag-related factors, are in the order of tens of meters per second or less, i.e., relatively low with respect to those generated by ejection pulses, reaching hundreds of meters per

second; and (ii) in contrast to rising pyroclasts, which are ejected from a relatively small area at relatively high volume concentrations, falling pyroclasts are more dispersed away from each other, reducing the probability of collisions. While the occurrence of ejection pulses in Plinian and hydromagmatic eruptions is highly plausible but less documented, their control on Strombolian and Vulcanian eruptions is clear [Scharff *et al.*, 2012; Gaudin *et al.*, 2017]. Future realistic modeling of VBP collisions during volcanic eruptions will have to incorporate the role of ejection pulses, following the way suggested by Tsunematsu *et al.* [2014].

The outcome of an in-flight collision between two VBPs is highly variable depending on the mechanical/rheological behavior of the projectiles, according to their temperature and chemical-mineralogical-textural composition. When eruptive temperature and pyroclast vesiculation are relatively low, as it is the case for blocks from Vulcanian eruptions, elastic response is observed to dominate the collision, and only rarely VBPs split up into a few, large fragments. During Strombolian eruptions, conversely, eruptive temperatures are typically higher and magmas more mafic (silica poor) in composition, and consequently pyroclasts tend to have a lower viscosity, at least before significant cooling lowers their temperature. Before that, a dominantly plastic behavior dominates, limiting elastic momentum transfer. In these cases, we observe that collision energy tends to dissipate by deformation of the bombs, rather than by changes in their momentum as a whole. A range of behaviors then emerges in colliding bombs, including merging, stretching, tearing, piercing, and all possible combinations. Collisions, in fact, appear to be a viable process to form very elongated bombs, as well as to accelerate the rate of spinning of colliding VBPs. When deformation is an important outcome, the changes in direction caused by the collision will be further complicated by shifts in the center of mass of the VBP as it attempts to readjusts its shape to the new, postcollision aerodynamic balance.

At Stromboli, previously observed collisions of bombs were modeled using elastic theory, although the possible presence of an inelastic component was acknowledged [Vanderkluisen *et al.*, 2012]. Almost all collisions we observed are inelastic, but it is important to note that our videos mostly focused on an area close to the eruptive vents, where significant bomb cooling was just beginning. Higher up, a different thermal, and hence mechanical behavior may result in a different collision outcome. In fact L. Spallanzani, at Stromboli in 1788 A.D., observed in-flight bomb collisions resulting in agglutination and fragmentation in the lower and higher portions of the eruptive jets, respectively, as a result of their rheological and mechanical properties (*"Diverse urtando insieme si spezzano, il che accade quando si trovano a certa elevatezza; ma in maggior vicinanza al vulcano, invece di rompersi, pel toccamento si conglutinano talvolta in una sola per la qualche liquidità che ritengono"*: many of them (bombs) hitting each other break down, and this happens at some elevation, but closer to the volcano, instead of breaking, by touching they agglutinate in a single one, due to some residual liquid component they still retain.) (L. Spallanzani, *Viaggi alle due Sicilie*, quoted in Mercalli, [1907], our translation). It is hard to assess the potential impact of these complex collision outcomes on the final landing position of VBPs, which is crucial for hazard assessment, but it seems likely that a simple elastic collision model may not be adequate to describe VBPs motion in Strombolian eruptions.

4.4. Ground Impact

As for VBPs in-flight fragmentation and collision, also the ground impact dynamics we observed bear consequences on the retrieval of eruptive conditions from the features of pyroclastic deposits. On slopes as steep as 45° or more, bombs are invariably fragmented by impact with the ground, and the original shape and size of fragments is further modified by rolling downward, by which process still molten fragments may agglutinate with each other and even incorporate previously deposited pyroclasts. The preservation of projectiles on impact with the ground is strongly dependent also on the properties of the landing substratum. Bombs and blocks from the same eruptive event have in fact been observed to survive on impact with loose pyroclastic deposits and fragment to various degrees on impact with lavas or boulders [cf. Fitzgerald *et al.*, 2014]. These observations are a compelling warning bell, once again telling us how much these factors must be kept in mind in interpreting the grain size and shape distributions of coarse-grained pyroclastic deposits.

Our parameterization of the impact ejecta fan suggests that impactor size may have a stronger control than kinetic energy on the diameter of the fan, proxy for crater size, at least within the observed variability of substratum characteristics and slope, impact angle, and VBP density. The kinetic energies we measured on impact match well with those calculated for the VBPs of the 2014 Ontake eruption [Tsunematsu *et al.*, 2016]. Using the dimensionless parameters defined in section 2.2.5, it is possible to estimate a scaling law

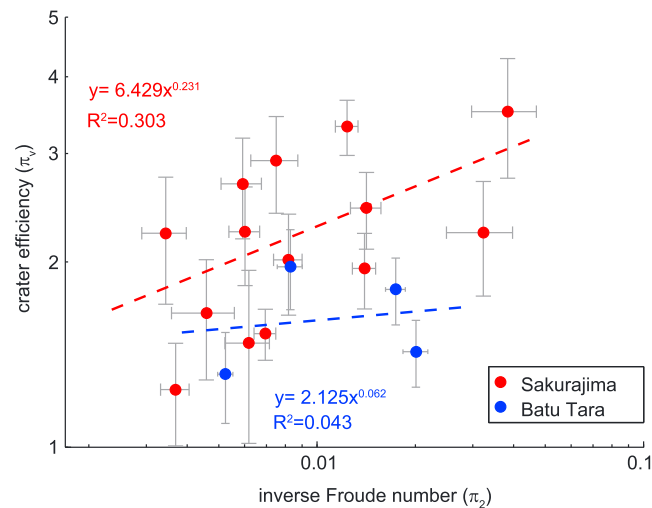


Figure 21. Relationship between the crater efficiency and the inverse Froude number of the observed impact craters (dashed lines are the corresponding power law fitting). The different slopes are reflecting differences in the impacted substratum.

relationship of impact crater formation from the high-speed video observations. As a first-order approximation, the crater efficiency (equation (42)) can be estimated as the D_c/D ratio, whereas the inverse Froude number (equation (43)) can be calculated directly from the parameters measured in the high-speed videos, starting from the raw data presented in Figure 15. For the measured impact craters from Batu Tara volcano the crater efficiency increases as the inverse Froude number increases according with the power law given by $\pi_v = 2.125\pi_2^{0.062}$. For Sakurajima volcano the scatter is larger, but in general π_v increases with higher values of π_2 , and the best power law fitting is obtained with $\pi_v = 6.429\pi_2^{0.231}$ (Figure 21). Since the yield strength of the substratum is not known, the rele-

levance of π_3 cannot be determined. However, we note that VBPs at Sakurajima, which typically erupts abundant fine-grained volcanic ash, probably impacted a substratum softer than that impacted by VBPs at Batu Tara, where large boulders and lava blocks are evident.

The velocity and spread angle of crater ejecta have been shown to be related to the scaled depth of chemical explosions [Housen and Holsapple, 2011; Ohba et al., 2002; Taddeucci et al., 2013]. In turn, the penetration depth of an impactor is also a function of its velocity, impact angle, and substratum properties [Manga et al., 2012, and references therein]. By combining the measured features of the crater ejecta with the size and velocity of the VBP it could be possible to link penetration depth and impact kinetic energy, using the video-derived information to obtain in situ information about the density of the VBPs, into a first-order function of their density and the characteristics of the substratum. The same substratum characteristics also control the bouncing of VBPs, extending further the related hazards, including their potential for starting wildfires in vegetated areas [e.g., Andronico et al., 2013]. It must be also considered that crater ejecta and fragments of broken projectiles represent a significant source of hazard in the vicinity of the landing site of a VBP [Fitzgerald et al., 2014]. Our observations reveal that, for impact velocities and kinetic energies in the 10–30 m s⁻¹ and 10²–10⁵ J ranges, respectively, fragments of the VBPs may reach distances in excess of 5 m from the original point of impact.

4.5. Trajectories

The centimeter-scale resolution of most of our videos allowed a detailed tracking of the apparent trajectory of VBPs. In turn, detailed tracking has unveiled how multiple in-flight processes cause the trajectories to deviate from the ideal, ballistic path.

The simplified case of a ballistic trajectory accounting only for gravity and drag would describe a smooth curve very similar in shape and velocity to that shown in Figure 16a. Deviations from a simplified case are larger and more common in the ascending part of trajectories than in the descending one. Three factors contribute to this asymmetry. The first factor is the higher concentration of VBPs close to the vent than at higher elevations, due to their proximity with their release source, i.e., the volcanic vent. Higher VBP concentration enhances the probability of trajectory-modifying collisions. It also enhances the possibility that the turbulent wake of preceding VBPs affects the following ones or even combines to alter the overall structure of the volcanic jet. The second factor is that the volcanic jet itself significantly affects the initial ascent of VBPs, as previously postulated and observed [e.g., Mastin, 2001; Taddeucci et al., 2015a]. Indeed, the gradual acceleration we observe for some ascending VBPs can be explained by their entrainment in a volcanic gas jet. Projectiles can be moved into the jet either after being pushed there by a collision (Figure 16l) or by

getting engulfed by a new jet from a subsequent ejection pulse (Figure 16k). The third factor is the higher temperature and lower viscosity of projectiles next to the vent in comparison with later stages of their trajectory. Lower projectile viscosity allows for easier deformation and tearing apart, which in turn change the VBP shape and aerodynamic response, eventually affecting its flight trajectory. In addition, both deformation and tearing will have a larger influence on appendages, asperities, and other flow-resistant parts of the projectiles, which are thus expected to offer lower resistance to air and a more stable trajectory later in their flight. All these effects are expected to increase the range of VBPs by effectively delaying their deceleration by the surrounding atmosphere.

During the descending fraction of their trajectory, VBPs deviations from an ideal ballistic case are mainly related to lateral winds and rotations. In particular, rotation of large, irregular projectiles causes both wobbling motions around their center of mass and fluctuations, even drastic, in their settling velocity.

4.6. Drag Force Acting on Flying Volcanic Ballistic Projectiles

Very few VBPs were observed to settle at a constant velocity (terminal fall velocity) due to their relatively large diameter and short travel path. However, it is still possible to estimate the drag force and the drag coefficient from the trajectory observations. In the case of the videos lasting less than 2 s, a constant C_d computed from the average acceleration yields a good fit (Figure 19) for most of the particles. However, the apparent C_d values calculated in this way reach up to 10 and down to close 0.1 (Figure 20) which largely extends the range of previous experimental data (up to 1.25, Figure 4), measurements made for a number of objects and shapes at $Re > 10^4$ (usually $C_d < 2$ [e.g., Hoerner, 1965]) and all the previously C_d values used in ballistic studies, up to 2 for supersonic conditions, section 2.2.2). These very high apparent C_d values can be the result of a number of factors, which might include some combination of the following.

1. As mentioned above, these apparent C_d were calculated considering the magnitude of the relative velocity between the VBP and air calculated only from the two observed (projected) components. If there is a significant component of the relative velocity perpendicular to the image plane, which is not considered, the magnitude of the relative velocity would be underestimated and the corresponding C_d would be overestimated. For instance, if for a given particle the component of the relative velocity vector perpendicular to the image plane has the same magnitude as the sum of the components observed in the image plane, then the apparent C_d would be greater than the actual C_d value by a factor of 2. This factor would be even higher if the VBPs are moving predominantly perpendicular to the image plane, although there is no reason to believe this to be the case. For this reason, the apparent C_d values we obtain should be considered as upper limits. A detailed analysis for the VBPs motion recorded with two different cameras [e.g., Gaudin *et al.*, 2016] will be very useful to evaluate this effect.
2. The experimentally derived model proposed by Bagheri and Bonadonna [2016] suggests that relatively high C_d values can be obtained for certain kN (Table 3) corresponding to particles with low flatness and elongation values (Figure 5). These shape factors cannot be measured from individual frames, but a detailed shape analysis throughout the trajectory of rotating particles around certain axis might allow a 3-D reconstruction of the particle shape and evaluate, whereas very high C_d values correspond to high kN as proposed by Bagheri and Bonadonna [2016]. In fact, most of the observed VBPs have very irregular shapes and high aspect ratios (Figure 8).
3. Most of the VBPs rotate significantly, although we tried to minimize this effect in the selected study cases. So far, this effect has been addressed simply by considering the average particle diameter. However, rotation may induce secondary motions that can change the drag force significantly. For instance, ductile bombs may deform in-flight due to rotation acquiring characteristic shapes. This deformation comes from the drag force. The drag coefficient of irregular rotating particles is still very poorly understood, but high-speed, high-resolution observations of rotating pyroclasts may provide new qualitative and quantitative information on this phenomenon.
4. Another significant difference between the volcanic projectiles and laboratory experiments is temperature. Pyroclasts can be ejected with temperatures above 600°C which may change the thermodynamic properties of the boundary layer (for instance, decreasing air density, increasing dynamic viscosity, and therefore reducing the Reynolds number) and may generate thermal instabilities and convection flows in the vicinity of the projectile, which in turn may affect the drag force. This topic is still very poorly understood.

Another interesting observation is that Figure 20 shows that apparent C_d decreases with Re . This decrease is more abrupt in the particles from Etna at $Re \sim 2\text{--}4 \times 10^5$, which might correspond to Re_{critical} . This value is very similar to that observed for smooth spheres (Figure 4) but 1 order of magnitude higher than Re_{critical} measured by *Alatorre-Ibargüengoitia and Delgado-Granados* [2006] for pyroclasts ($Re_{\text{critical}} \sim 2 \times 10^4$). As mentioned before, surface roughness decreases Re_{critical} . Another explanation may come from the temperature, considering that high-temperature pyroclasts may reduce the effective Re in the boundary layer, thus increasing the apparent value of Re_{critical} . Further analysis of particles moving at higher Re is needed to estimate C_d values for such conditions.

Figure 20 also shows that there is a significant scatter in the observed C_d values that is not only caused by the uncertainties associated with the measurements but also reflects the variability in the aerodynamic behavior of the VBPs. Calculated high C_d values match the suggestion given by *Taddeucci et al.* [2015a] and *de' Michieli Vitturi et al.* [2010] who signaled that C_d values >1 are required to reproduce correctly the in-flight behavior of centimeter-sized pyroclasts. On the other hand, we also calculated low C_d values that resemble C_d s for spheres at $Re > Re_{\text{critical}}$, which have been invoked to calculate reasonable ejection velocities for VBPs observed at large distances from the vent [e.g., *Waite et al.*, 1995]. These low C_d values might correspond to pyroclasts with shapes and textures more aerodynamic which allow them to travel farther distances and therefore have a particular interest for hazard assessment.

5. Conclusions

By combining an extended review of the literature with original, high-speed observations and parameterizations of VBPs, we can now draw the following general conclusions.

1. The commonly observed in-flight tearing and fragmentation of bombs, together with their common breaking on impact with the ground, have important implications for the retrieval of eruption parameters from the grain size distribution of pyroclastic deposits. Our observations lead us to conclude that the grain size distribution of proximal (near vent) deposits of mafic composition (silica-poor and less viscous magma) will invariably be finer than that originated at the vent. Even in rare cases where projectiles may survive ground impact, the larger ones will not survive in-flight fragmentation, which represents a so far largely neglected factor controlling the grain size distribution of pyroclastic deposits.
2. In-flight collision between VBPs is common and its outcomes are greatly variable, including changes in the size, shape, flight direction, and rotation rate of the projectiles. The dominance of one outcome over another is a complex function of the VBP properties (ductile bomb versus brittle block) and also of the location and timing of the collision (early, near-vent collisions or later, higher-elevation ones). The timing, frequency, and location of collisions are entirely controlled by the occurrence and specific features (e.g., frequency, velocity, and direction) of the multiple ejection pulses that combine in any single volcanic explosion.
3. Rotation and spinning are ubiquitous features of VBPs, with strong effects on their travel path. Rotations are often complex and time-space variable and are enhanced by collisions. Spinning is also common, and the Magnus effect, previously invoked as a possible explanation for VBPs deviating trajectories, has now been demonstrated to be able to play an important role.
4. We offer new, observation-based relationships between the velocity and size of VBPs on impact with the ground and the size of the resulting impact crater. These relationships outline the relevance and variability of the characteristics and strength of the impacted substratum and represent the first field-derived estimates of impact craters during volcanic eruptions.
5. The ascending trajectory of VBPs is influenced not only by collisions, rotations, deformation, and fragmentation but also by the gas stream leaving the vent. While previous work has shown drag reduction near the vent, here we show how VBPs may actually be reentrained and accelerated within the stream above the vent.
6. New apparent drag coefficient estimates yield higher than expected values. Even if recent observational and numerical studies have suggested relatively high drag coefficient values, the observation-based estimates presented here span a larger range than the values considered in previous ballistic studies. We hypothesize that the critical Reynolds number may be affected by the ballistic temperature via

changes in the thermodynamic properties of the boundary layer, and therefore, the drag coefficient may be reduced at higher velocities comparing with data at ambient temperature. The high values we obtain may arise from a number of factors, both instrumental (lack of three-dimension characterization of the trajectories and lack of more sophisticated description of VBP shape) and general, including a limited knowledge of the aerodynamic behavior of large, very irregular, rotating, hot bodies traveling at relatively high Reynolds numbers. All these considerations call for more observational, experimental, and theoretical studies on the subject.

In conclusion, our review highlights a disparity in our current knowledge of volcanic ballistic projectiles: while field data get more and more abundant, current theoretical models for ballistic transport can be very sophisticated in certain aspects that are sometimes secondary, while neglecting large uncertainties on first-order processes such as rotation, drag coefficient, interactions with the gas jet, collisions, and in-flight deformation. This is because the input data for models are currently disproportionately rough in comparison to computational capabilities, pointing to the need for additional experimental modeling, deposit analysis, and direct observation.

The high-speed observations we presented are just a first example of the potential for precise video observations in providing invaluable information for parameterization of different process and model calibration and validation. More and more systematic observations are required and expected in the future, particularly for higher-intensity volcanic eruptions not covered in this study. It is our hope that the new perspectives we offered may help improving future projectile fallout hazard mapping and forecasting and shed light on the interpretation of projectile deposits from past eruptions, both on Earth and on other planets.

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